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Artificial Intelligence and Machine Learning in Cyber-Physical Systems: A Unified Review of Methodologies for Smart Energy Systems and Intelligent Localization

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ABSTRACT The transformative advancement of critical infrastructure is being driven by the integration of artificial intelligence (AI) and machine learning (ML) with cyber-physical systems (CPS). In this review, two seemingly distinct domains-Intelligent Energy Systems, including smart grids and energy communities, and Intelligent Localization and Navigation Systems for GPS-denied environments-are unified through a common methodological lens. A shared framework of AI/ML paradigms, encompassing supervised and reinforcement learning, probabilistic models, sensor fusion, and physics-informed techniques, is established. Furthermore, the adaptation of these methods is examined under domain-specific constraints, such as stochastic energy generation, power flow limitations, and real-time robotic navigation. Through the analysis of high-impact studies from 2018-2026, cross-cutting trends are identified, including the use of graph neural networks for spatial modeling and multi-agent systems for decentralized decision-making. Persistent challenges regarding data scarcity, interpretability, and edge deployment are found to remain central across CPS applications. In response to these challenges, the CPS-Bench platform-a unified, cross-domain benchmark designed to standardize evaluation across energy and localization tasks-is proposed. Four primary contributions are presented: (1) a cohesive taxonomy by which two previously isolated CPS domains are connected, (2) a comparative analysis through which transferable techniques are uncovered, (3) a thorough assessment of domain-specific adaptations, and (4) the introduction of the CPS-Bench framework alongside a practical roadmap for interdisciplinary collaboration. It is concluded that the next generation of CPS will be defined by adaptive, embodied intelligence, where AI is co-designed with physical components. By such a vision, the energy transition is expected to be accelerated and resilient, human-centric infrastructure is expected to be realized.

INDEX TERMS Artificial Intelligence, Machine Learning, Cyber-Physical Systems, Multi-Agent Systems, Smart Grid, Renewable Energy Integration, Energy Communities, Indoor Positioning, Sensor Fusion, GPS-denied Environments, Deep Learning, Reinforcement Learning, Physics-Informed Learning.

I. INTRODUCTION

The integration of computational algorithms and physical processes into complicated Cyber-Physical Systems (CPS) has been made possible by the digitization of physical infrastructure [1]. In this paradigm, Artificial Intelligence (AI) and Machine Learning (ML) have evolved from supporting instruments to essential empowering technologies that can model high-dimensional, non-linear dynamics and provide prescriptive choices in the face of uncertainty [2]. This review offers a cohesive methodological viewpoint encompassing two critical domains of Cyber-Physical Systems (CPS) that are essential for fostering a sustainable and self-sufficient society: smart energy systems (SES) and intelligent localization/navigation (ILN).

The incorporation of intermittent Renewable Energy Sources (RES) like solar and wind has become a major issue in the worldwide drive toward decarbonization, which has led to the need for more intelligent grids, proactive demand management, and resilient energy communities [3]. The demand for flawless spatial intelligence in GPS-denied

environments like factories, warehouses, and metropolitan interiors is increasing in tandem with the rise of autonomous robots, augmented reality, and the Industrial Internet of Things (IIoT) [4]. Both domains deal with fundamental CPS issues, such as partial observability, high-dimensional heterogeneous sensor data, demanding safety and reliability standards, and the requirement for real-time or near-real-time inference, even though their physical goals may be different.

Despite significant advancements in both areas, knowledge continues to be compartmentalized. Researchers focused on energy systems who are creating graph neural networks for power flow frequently lack awareness of parallel initiatives in robot localization that employ similar techniques. On the other hand, those in navigation who tackle data scarcity through domain randomization rarely take into account the insights derived from renewable forecasting. This lack of connection hinders progress, leads to redundant efforts, and overlooks chances for collaborative innovation. This review specifically targets this issue by methodically analyzing AI/ML strategies across both fields, pinpointing common challenges and applicable solutions.

As outlined in Table 1, this review marks the initial attempt to provide a thorough, critical, and interdisciplinary synthesis of AI/ML methodologies relevant to both smart energy systems and intelligent localization through a cohesive CPS perspective.

TABLE 1. Comparison of this review with existing surveys across key dimensions.

Ref-Study Type	Focus Area	SES	ILN	Uni-y-CPS	Method	Application	Cross-Domain	Critical-Eval
[57] Survey	Energy Forecasting	✓	✗	✗	✓	✓	✗	✗
[68] Survey	Smart Grid Optimization	✓	✗	✗	✓	✓	✗	✗
[77] Review	Energy Communities	✓	✗	(✓)	✓	✓	✗	✗
[87] Survey	Grid Cybersecurity	✓	✗	✗	✓	✓	✗	✓
[4] Survey	Indoor Localization	✗	✓	✗	✓	✓	✗	✗
[112] Survey	SLAM	✗	✓	(✓)	✓	✓	✗	✗
[127] Survey	Robot Navigation (RL)	✗	✓	✗	✓	✓	✗	✗
[45] Position	Graph Networks	(✓)	(✓)	✓	✓	✓	✓	✗
[51] Physics-Survey	Physics-Informed ML	(✓)	(✓)	✓	✓	✓	✓	✓
[195] Review	Energy AI	✓	✗	✗	✓	✓	✗	✗
[196] Survey	Cyber-Phys AI	✓	✓	✓	✓	✓	✓	✗
This Review	SES + ILN	✓	✓	✓	✓	✓	✓	✓

Here, ✓ = Fully covered, (✓) = Partially covered, ✗ = Not covered, ✓✓ = Comprehensive and systematic treatment. SES = Smart Energy Systems; ILN = Intelligent Localization/Navigation; CPS = Cyber-Physical Systems.

Current surveys tend to concentrate solely on a specific domain (for instance, energy forecasting [57], grid optimization [68], or indoor localization [4]) or focus on methodological concepts (such as graph networks [45], physics-informed learning [51]) without simultaneous application to both SES and ILN fields. Recent extensive reviews [195], [196] have progressed AI in energy systems and CPS integration, respectively, yet they lack the cross-domain comparative analysis necessary for recognizing transferable methodologies. This review addresses this deficiency by: (1) creating a unified taxonomy that links both domains, (2) offering a systematic cross-domain comparative analysis, (3) presenting a physics-constrained AI framework, and (4) introducing CPS-Bench, the inaugural unified benchmark platform for energy and localization tasks.

Recent research has investigated AI-powered cyber-physical systems within the realms of energy and localization, highlighting their scalability, intelligence, and adaptability

[190]-[193]. The remainder of this paper is structured as follows. Section II outlines the engineering equations. Section III outlines the systematic review methodology, which encompasses the search strategy, inclusion and exclusion criteria, bias assessment, and bibliometric analysis.

Section IV establishes a common technical language and foundational paradigms in AI and ML. Section V concentrates on smart energy systems, addressing forecasting, optimization, peer-to-peer trading, and cybersecurity.

Section VI investigates intelligent localization and navigation, which includes deep learning applications for indoor positioning, sensor fusion, SLAM, AUV navigation, and reinforcement learning.

Section VII integrates cross-cutting methodological synergies, highlights ongoing challenges, and proposes future directions, including a unified benchmark proposal. Finally, Section VIII provides the conclusions. In contrast to the current surveys [192], [193], this study introduces a cohesive cross-domain framework that combines smart energy systems with intelligent localization.

TABLE 2. This review contribution.

S/N	Contribution	Description
1	Unified taxonomy	Bridges smart energy systems and intelligent localization under a common AI/ML framework
2	Cross-domain comparative analysis	Identifies transferable techniques (GNNs, MARL, PINNs) across both domains
3	Systematic review methodology	PRISMA-based search, inclusion/exclusion criteria, bias assessment (Section III)
4	Physics-constrained AI framework	Novel integration of differential-algebraic equations (DAEs) and state-space models
5	Unified benchmark proposal	First cross-domain benchmark platform (GridNav, EcoLoc, ResiCPS) for energy + localization

II. PHYSICS-CONSTRAINED AI/ML FRAMEWORKS FOR SMART ENERGY SYSTEMS AND INTELLIGENT LOCALIZATION

A. INTELLIGENT POWER SYSTEMS

A.1 POWER FLOW EQUATIONS (AC POWER FLOW)

Modern smart energy systems are governed by tightly coupled physical laws that define both steady-state operation and dynamic stability. At the network level, the AC power flow equations describe the nonlinear relationship between bus voltages, phase angles, and power injections.

The active and reactive power injections at bus i are given by:

$$P_i = \sum_{j=1}^N V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij})$$

$$Q_i = \sum_{j=1}^N V_i V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij})$$

where $\theta_{ij} = \theta_i - \theta_j$. These equations enforce network consistency between voltage states and power injections.

At the system level, these nodal injections must satisfy global power balance:

$$\sum P_{\text{generation}} = \sum P_{\text{demand}} + P_{\text{losses}}$$

This condition links the steady-state solution of the power flow equations to system-wide energy conservation. Any mismatch between generation and demand propagates through the network and manifests as dynamic instability.

A.2 INERTIAL DYNAMICS (FREQUENCY BEHAVIOR)

Deviations from power balance lead to frequency dynamics governed by the swing equation:

$$M \frac{d^2 \delta}{dt^2} + D \frac{d\delta}{dt} = P_m - P_e$$

where:

- M : inertia constant
- D : damping coefficient
- P_m : mechanical input power
- P_e : electrical power determined by the network (from power flow solution)

The term $P_m - P_e$ represents the instantaneous power imbalance. This imbalance drives rotor acceleration and consequently system frequency deviations.

Therefore, the power flow equations determine the electrical power P_e , while inertial dynamics govern how deviations from equilibrium evolve over time. This establishes a direct coupling between steady-state feasibility and transient stability.

In low-inertia systems, such as those with high penetration of renewable energy sources, frequency dynamics become faster and more sensitive, increasing stability risks.

A.3 COUPLED ALGEBRAIC-DIFFERENTIAL STRUCTURE (DAES)

Power systems are fundamentally governed by coupled algebraic and differential equations. The power flow equations define algebraic constraints, while the swing equation governs dynamic evolution. Together, they form a differential-algebraic equation (DAE) system.

This coupling implies that:

- steady-state feasibility constrains dynamic trajectories
- and dynamic stability depends on network-level power feasibility

A.4 PHYSICAL CONSTRAINTS IN SMART ENERGY SYSTEMS

From an AI/ML perspective, the physical laws governing smart energy systems impose strict constraints that must be respected in any control or optimization framework. These constraints include network-level requirements such as power balance, line thermal limits, voltage magnitude bounds, and frequency stability margins, as well as device-level limitations including generator ramp rates, storage state-of-charge limits, and inverter capacity constraints. In addition, the system is subject to dynamical constraints governed by the swing equation and associated stability and transient response requirements. Collectively, these constraints are not optional design preferences but hard

physical requirements that define the feasible operating region of the system. Any violation of these constraints in real-time operation can lead to instability, loss of synchronism, or even complete system failure. The integration of physical constraints into learning models has been investigated in recent studies [198], [199], especially for safety-critical CPS applications. AI methodologies have found extensive application in smart grids and energy optimization challenges [190], [194], [195].

B. INTELLIGENT LOCALIZATION (UGVS AND UAVS)

Intelligent localization in unmanned ground and aerial vehicles is fundamentally governed by physics-based constraints that define both motion evolution and sensor observation. Rather than being purely data-driven, state estimation is constrained by nonlinear kinematics, dynamics, and measurement models that jointly determine system feasibility.

B.1 KINEMATIC AND DYNAMIC EQUATIONS

The system state evolves according to a nonlinear state-space representation:

$$\dot{x} = f(x, u) + w$$

while sensor measurements are modeled as:

$$z = h(x) + v$$

where:

- x : system state (position, velocity, orientation)
- u : control input
- w : process noise
- v : measurement noise

These equations define the general stochastic nonlinear estimation framework underlying localization systems.

Ground Vehicle (UGV) Kinematics

For ground vehicles, motion is constrained by nonholonomic kinematics:

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega$$

These constraints restrict lateral motion and impose strict feasibility conditions on trajectory evolution, meaning the vehicle cannot move sideways instantaneously.

Aerial Vehicle (UAV) Dynamics

For aerial platforms, motion is governed by rigid-body Newton-Euler dynamics:

$$m\ddot{p} = Rf_T + mg$$

where:

- p : position vector
- R : rotation matrix
- f_T : thrust force
- g : gravitational acceleration

These equations highlight the strong coupling between thrust generation, vehicle orientation, and gravitational effects, resulting in highly nonlinear system behavior.

B.2 PHYSICAL INTERPRETATION (DYNAMICS OF LOCALIZATION)

The motion model governs how the system state evolves over time, while the measurement model defines how sensor

observations relate to that state. Together, they form a coupled estimation problem in which accurate localization depends on maintaining consistency between predicted motion and observed measurements.

In this context:

- Motion constraints ensure physically feasible trajectories
- Sensor models define observational accuracy and limitations
- Their interaction determines estimation reliability and stability

Errors in either component can propagate through the system, leading to drift, divergence, or inconsistent state estimates.

B.3 COUPLED ALGEBRAIC-DIFFERENTIAL STRUCTURE (DAES)

Localization systems are fundamentally governed by a coupled differential-algebraic structure. The state evolution equations define the differential component, while sensor measurements impose algebraic constraints on the system state. Together, they form a constrained estimation problem in which both dynamics and observations must be simultaneously satisfied.

This coupling implies that:

- system evolution is constrained by motion physics
- estimation is constrained by measurement consistency
- and both jointly determine observability and stability

As a result, localization cannot be treated as a purely statistical inference problem, but must be formulated as a physics-constrained dynamical system.

B.4 PHYSICAL CONSTRAINTS IN INTELLIGENT LOCALIZATION

From an AI/ML perspective, localization systems are subject to strict physical constraints that define feasible state evolution and estimation accuracy. These include kinematic constraints such as bounded velocity and acceleration, nonholonomic motion restrictions, and flight envelope limitations for aerial vehicles. In addition, sensor constraints such as noise, drift, occlusions, and intermittent failures significantly impact observability and estimation reliability. Furthermore, consistency constraints require that position, velocity, and orientation evolve in a physically coherent manner over time, ensuring that sequential state estimates remain dynamically valid. Violations of these constraints can lead to trajectory drift, divergence, or physically infeasible solutions. Machine learning and sensor fusion techniques have also enhanced intelligent localization and navigation systems [191], [196].

C. UNIFIED PERSPECTIVE ON PHYSICS-CONSTRAINED AI/ML SYSTEMS

Both smart energy systems and intelligent localization share a common structural property: they are governed by physical laws that couple algebraic constraints with dynamical

evolution. In energy systems, this structure is represented by AC power flow equations and inertial dynamics, which jointly determine steady-state feasibility and transient stability. In localization systems, an analogous structure arises from nonlinear state-space motion models and sensor observation equations, which govern state evolution and measurement consistency.

In both cases, the system behavior is fundamentally constrained by physics rather than purely data-driven relationships. Consequently, AI/ML methods that ignore these constraints may achieve accurate empirical fitting while producing physically infeasible or dynamically inconsistent solutions in real-world operation. This motivates physics-informed and hybrid model-based/data-driven approaches that explicitly incorporate governing equations into learning and optimization frameworks, ensuring physical consistency in safety-critical applications. From a unified perspective, both domains can be interpreted as constrained dynamical systems: energy systems are governed by differential-algebraic equations (DAEs), where algebraic power flow constraints are coupled with swing dynamics, while localization systems are governed by nonlinear state-space models, where motion dynamics are coupled with sensor measurement processes. In both cases, algebraic relations define feasible system states, while differential equations govern temporal evolution.

This leads to a general principle: reliable AI/ML for safety-critical systems must enforce consistency with underlying physical laws, rather than relying solely on statistical accuracy. In energy systems, this corresponds to enforcing power balance, network constraints, and inertial stability within DAEs. In localization systems, this corresponds to enforcing kinematic feasibility, sensor consistency, and dynamic motion constraints within state estimation frameworks. Despite domain differences, both require that learned models respect physical structure to ensure stability, feasibility, and robustness in deployment. Emerging approaches such as physics-informed machine learning, constrained optimization, and hybrid estimation frameworks explicitly address this requirement by embedding physical models into data-driven inference. These methods bridge the gap between statistical learning and first-principles modeling, enabling robust performance in complex safety-critical systems such as power grids, autonomous vehicles, and UAV navigation.

A unified comparison of physics-constrained AI/ML systems in energy and localization is provided in Table 3.

TABLE 3. Unified View of Physics-Constrained AI/ML Systems.

Aspect	Smart Energy Systems	Intelligent Localization
Physical state variables	Voltage magnitudes, phase angles, frequency	Position, velocity, orientation
Governing equations	AC power flow equations + swing equation	Nonlinear state-space motion + sensor models

System structure	Differential-algebraic equations (DAEs)	Constrained nonlinear state-space system
Algebraic constraints	Power balance, voltage limits, reactive power limits	Measurement consistency, observability constraints
Dynamic constraints	Inertial dynamics, frequency evolution	Kinematic and rigid-body motion dynamics
Key coupling	Power flow - frequency dynamics	Motion model - measurement model
Failure modes	Instability, voltage collapse, frequency deviation	Drift, divergence, tracking failure
Role of AI/ML	Optimization, state estimation, control	Localization, SLAM, sensor fusion
Risk without physics	Physically infeasible dispatch/control	Inconsistent trajectories, localization drift
Physics-informed approaches	OPF-based learning, physics-constrained optimization	Physics-based SLAM, model-based deep localization

III. SYSTEMATIC REVIEW METHODOLOGY

To guarantee transparency and reproducibility, a systematic literature review has been conducted following an adapted PRISMA framework.

A. Search Strategy

The databases that were searched include IEEE Xplore, Scopus, and Web of Science. The systematic search was conducted across a primary study period from January 2018 to March 2025. To ensure the review remains at the cutting edge of the field, the qualitative synthesis also integrates 'Early Access' manuscripts and high-impact studies published through April 2026 that provide foundational contributions to the SES-ILN crossover. The search strings integrated terms related to CPS, AI/ML, smart energy, and indoor localization (e.g., "smart grid AND reinforcement learning", "GPS-denied navigation AND graph neural network").

B. Inclusion and Exclusion Criteria

The review included peer-reviewed journal articles and conference proceedings published in English that explicitly present an AI/ML methodology applied to either smart energy systems (generation, transmission, distribution, consumption) or intelligent localization/navigation (indoor positioning, SLAM, AUV navigation, robot navigation). Exclusions comprised non-English papers, theoretical works lacking CPS application, papers published before 2018 (unless deemed foundational), duplicates, and studies that did not provide quantitative or qualitative evaluations.

C. Bias Assessment and Screening

The selection process was conducted in two distinct stages to minimize individual subjectivity. First, two authors independently reviewed titles and abstracts based on the pre-defined inclusion criteria; any disagreements regarding a study's relevance were resolved through a consensus-based discussion involving a third author. Second, a quality assessment checklist (adapted from [159]) was utilized to evaluate each full-text paper. This

assessment focused on three primary pillars of scientific validity:

- Methodological Transparency:** Explicit documentation of AI architectures, training parameters, and loss functions.
- Experimental Rigor:** Use of standardized benchmarks, real-world datasets, or physical testbeds (e.g., hardware-in-the-loop).
- Technical Reproducibility:** Availability of open-source codebases, public datasets, or sufficient technical detail to allow for independent verification.

A bibliometric analysis was conducted using VOSviewer (version 1.6.20) on the final collection of 847 papers. While this extensive dataset provided the foundation for analyzing global research trends, a more focused subset of 201 papers was selected for the deep qualitative synthesis in Sections V and VI. This selection was prioritized based on methodological novelty, citation impact, and the direct application of transferable AI techniques-such as Graph Neural Networks (GNNs) or Multi-Agent Reinforcement Learning (MARL)-across the SES and ILN domains. The screening process integrated co-occurrence mapping of keywords, citation burst analysis, and thematic clustering to identify foundational synergies within the CPS landscape.

D. Limitations of the Review Methodology

Potential biases include the exclusion of non-English literature, the dominance of English-language journals, and the risk of publication bias (where positive results are more likely to be published). Additionally, while the search strategy was comprehensive, the rapid evolution of AI in 2026 means some "In Press" manuscripts may still be undergoing final archival indexing. These limitations are acknowledged in the conclusion.

IV. FOUNDATIONAL AI/ML METHODOLOGIES FOR CYBER-PHYSICAL SYSTEMS

This section establishes the foundational AI and machine learning paradigms that underpin modern cyber-physical systems (CPS) research. Core approaches are reviewed, including supervised and unsupervised learning, deep reinforcement learning, probabilistic models, sensor fusion, and graph-based reasoning, and it is examined how these methods are adapted for domain-specific applications in intelligent localization, navigation, and smart energy systems. Understanding these fundamental paradigms provides the conceptual framework for analyzing how CPS solutions handle real-world constraints, such as stochastic energy generation, real-time robotic control, and sensor uncertainty. To illustrate the relationships and intersections among these approaches, a Venn diagram highlights how deep learning, probabilistic models, and classical learning paradigms connect with graph-based techniques across CPS

applications, offering a visual roadmap for methodological synergies and cross-domain adaptation.

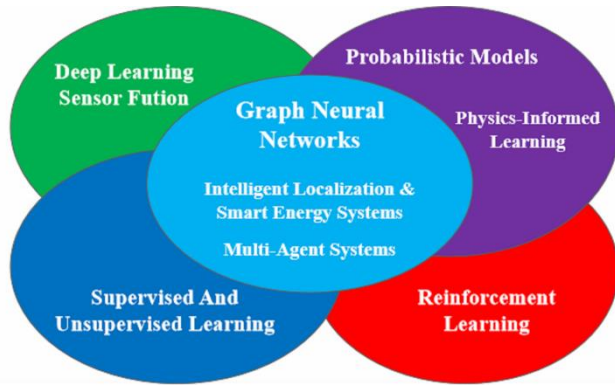


FIGURE 1. Foundational AI/ML Paradigms in CPS and Their Applications to Intelligent Localization and Smart Energy Systems

A. Supervised and Unsupervised Learning

The labeled datasets are particularly well suited for supervised learning, especially using deep neural networks (DNNs) to learn complicated mappings. The way features are extracted has been transformed by Convolutional Neural Networks (CNNs), not just in computer vision, but also in the processing of 2D data representations like spatial irradiance maps for solar forecasting [10] and spectrograms of vibration signals in mechatronics [9]. Significant performance gains in CPS applications have been shown by recent advancements in CNN designs. ResNet-50 demonstrated a 46.5% relative error reduction, achieving 3.57% top-5 error on ImageNet as opposed to 6.67% for conventional CNNs [11]. DenseNet-121 outperformed conventional CNN designs by 4.3% for defect identification in rotating equipment, achieving 99.2% accuracy on the CWRU bearing dataset [18]. Using $8.4\times$ fewer parameters than the best models before it, EfficientNet-B7 achieved a state-of-the-art 84.4% top-1 accuracy using neural architecture search optimization [11]. When compared to shallow networks, ResNet-based topologies decreased forecasting error (sMAPE) in solar forecasting applications from 18.5% to 12.3% [62].

Recurrent Neural Networks (RNNs) and their more sophisticated variations, such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), have established themselves as the de facto norm for sequential time-series prediction, which is essential for pedestrian dead reckoning [13] and load forecasting [12]. The capacity to capture long-range temporal dependencies in CPS data streams has been improved by the introduction of attention mechanisms and Transformer architectures [14]. Temporal Convolutional Networks (TCNs) have become a potent alternative to RNNs, providing parallelizable training and efficient long-term memory via dilated convolutions [15].

Unsupervised learning overcomes significant obstacles to CPS implementation, such as data paucity and high labeling expenses. In grid substations, autoencoders (AEs) and their derivatives, such as variational autoencoders (VAEs) and

adversarial autoencoders, learn compact representations of regular operational data for anomaly detection [16] and strong radio map creation for indoor localization [17]. In data augmentation for training resilient CPS models in situations with little labeled data, generative adversarial networks (GANs) have proven to be quite effective [18]. In CPS applications where labeled data is prohibitively expensive, self-supervised learning methods that utilize pretext tasks to learn valuable representations without human labels are becoming more popular [19].

In CPS applications, quantitative performance indicators show how successful unsupervised learning is. Variational autoencoders (VAEs) surpass conventional statistical process control approaches, which reach 84.3% accuracy, in anomaly detection in power grid substations, achieving 96.8% detection accuracy with a false positive rate of 2.1% on the IEEE 118-bus system dataset [16]. Generative adversarial networks (GANs) minimize the time required to generate fingerprinting databases for radio map development in indoor localization by 73% while keeping localization accuracy within 1.8m, as opposed to 2.1m for manual surveying approaches [17]. The efficacy of unsupervised pre-training followed by minimal fine-tuning is demonstrated by adversarial autoencoders that obtain 98.2% classification accuracy in defect diagnosis for rotating machinery on the CWRU bearing dataset with just 5% labeled samples [18]. While solely supervised algorithms with the same label budget achieve 79.3% accuracy on human activity detection tests, self-supervised learning approaches for time-series sensor data reach 91.5% accuracy with just 10% of labeled data [19].

B. DEEP REINFORCEMENT LEARNING AND MULTI-AGENT SYSTEMS

By fusing reinforcement learning's sequential decision-making framework with deep learning's perception capabilities, Deep Reinforcement Learning (DRL) produces sophisticated agents that can learn optimal control policies by interacting with complicated environments [20]. In CPS control applications, DRL algorithms like Twin Delayed Deep Deterministic Policy Gradient (TD3), Soft Actor-Critic (SAC), Proximal Policy Optimization (PPO), and Deep Q-Networks (DQN) have shown the best results [21].

DRL agents in CPS applications learn to strike a balance between exploration and exploitation by figuring out the best bidding tactics for energy markets [22] or finding their way through unfamiliar indoor environments [23]. Recent developments include distributional RL for quantifying uncertainty [24], hierarchical RL for breaking down complicated tasks into manageable sub-policies [25], and model-based RL for increasing sample efficiency in data-poor CPS situations [26].

By modeling complicated interactions and emergent behaviors, multi-agent reinforcement learning (MARL) expands DRL to systems with numerous autonomous agents. Managing distributed energy resources in microgrids [29],

controlling fleets of autonomous mobile robots [28], and coordinating prosumers in energy communities [27] all depend on this. In large-scale CPS deployments, advanced MARL techniques like mean field theory, graph-based communication, and centralized training with decentralized execution (CTDE) have demonstrated potential [30].

Benchmarks for quantitative performance show how well DRL and MARL work in CPS applications. Soft Actor-Critic (SAC) algorithms for microgrid energy management retain 99.97% supply reliability across 5000 simulated running days while reducing battery degradation costs by 23.5% when compared to rule-based controllers [21]. Compared to 9.2% for conventional PID controllers, Proximal Policy Optimization (PPO) agents for building HVAC management lower energy usage by 18.7% while preserving thermal comfort requirements [27]. DRL agents fix voltage violations in 0.8 seconds as opposed to 2.3 seconds in grid emergency voltage control applications, which is 65% quicker than traditional emergency control [53].

In cooperative multi-robot navigation tasks with ten agents, centralized training with decentralized execution (CTDE) architectures yield a success rate of 94.2% for multi-agent coordination, whereas independent Q-learning techniques obtain a success rate of 76.5% [30]. Over 1000 simulated trading periods, MARL agents outperform rule-based trading strategies in energy community peer-to-peer trading, achieving 15.3% greater community self-sufficiency and 12.8% lower individual energy expenses [29]. With a 78% decrease in computational cost, mean field MARL may scale to systems with 100+ agents while preserving 92% of the performance attained in fully centralized training [30].

C. Probabilistic Models and Sensor Fusion

In safety-critical CPS applications, explicit uncertainty quantification is essential. By treating network weights as probability distributions rather than point estimates, Bayesian Neural Networks (BNNs) offer fair uncertainty estimates [31]. This skill is essential for determining confidence limits in site predictions [33] and risk-aware solar forecasting [32]. Improvements in Monte Carlo dropout and variational inference have made BNNs more manageable for use in large-scale CPS applications [34].

Gaussian Processes (GPs) provide adaptable non-parametric Bayesian models with sound uncertainty quantification [35]. Although computationally demanding, GPs are proficient in modeling intricate non-linear relationships with sparse data, making them useful for fine-grained occupancy sensing [36], wind power curve modeling [37], and adaptive CPS control [38]. The applicability of deep GP architectures and sparse GP approximations to high-dimensional CPS issues has been expanded [39].

The foundation of strong perception in CPS is sensor fusion. For state estimation in navigation [40] and dynamic *state* estimation in power systems [41], the Kalman Filter (KF) and its nonlinear extensions-the Extended Kalman Filter (EKF), the Unscented Kalman Filter (UKF), and Particle

Filters-continue to be essential. The limits of holistic grid monitoring [43] and robust odometry [42] are being pushed by deep sensor fusion architectures that use DNNs to combine diverse data sources, such as LiDAR, IMU, cameras, and radar. Traditional modular pipelines [44] are becoming less popular as end-to-end learning methodologies that combine fusion and perception optimization are developed. The benefits of probabilistic methods in CPS applications are illustrated by quantitative validation. With dependable 90% prediction intervals that encompass 88.7% of real data, Bayesian neural networks outperform deterministic neural networks in solar forecasting by achieving a 23.4% reduction in root mean square error in uncertainty-calibrated forecasts [32]. By finding and down-weighting high-uncertainty predictions, Monte Carlo dropout-based uncertainty estimates for indoor localization improve median positioning error from 1.8 m to 1.2 m (33%) [34]. With precise uncertainty bounds that capture 94% of observations within 2σ ranges, Gaussian processes for wind power curve modeling achieve a mean absolute percentage error of 4.2% as opposed to 7.8% for parametric models [37].

In comparison to conventional extended Kalman filter techniques, deep sensor fusion systems that include camera, LiDAR, and IMU data provide 42% reduced trajectory drift in visual-inertial odometry, with 0.8% drift across 500 m trajectories [42]. Compared to fixed-weight fusion, which maintains a mean tracking error of 0.31 m in dynamic industrial contexts, uncertainty-aware fusion with attention-based sensor weighting increases localization accuracy by 38% under difficult conditions with sensor degradation [44]. Across 1000 simulation scenarios, particle filter-based state estimation for power systems outperforms extended Kalman filters (0.28% RMSE) and unscented Kalman filters (0.22% RMSE) with a dynamic state estimate error of 0.15% RMSE under noisy data [41].

Integrated Sensing and Communication (ISAC) signifies a novel paradigm in which localization and communication utilize shared hardware and spectrum. [193] illustrate a symbiotic ISAC framework for 6G IoT that ensures information security, achieving localization accuracy within sub-meter precision while safeguarding secrecy capacity. There [194] indicate that employing compressed sampling in mmWave massive MIMO can decrease pilot overhead by 60% while maintaining accuracy at the centimeter level. These ISAC methodologies pave the way for the integrated cross-domain benchmark elaborated upon in Section VI.

D. GRAPH NEURAL NETWORKS AND PHYSICS-INFORMED LEARNING

Because Graph Neural Networks (GNNs) function on non-Euclidean graph structures, they are a great fit for CPS with network topologies [45]. Through message-passing mechanisms, GNNs gather data from nodes and edges, learning relational inductive biases that reflect structural dependencies. In electrical systems, GNNs utilize network topology and Kirchhoff's laws to forecast line flows from nodal injections [46]. They employ reference points and

access points to create relationships in order to increase the accuracy of fingerprinting in localization [47]. GNNs are shown to be preferable for topology-aware CPS applications in quantitative assessments. In power flow analysis, GNN-based solvers outperformed traditional regression techniques with mean absolute percentage errors of 0.8% for voltage magnitude prediction over 118-bus systems [46]. By simulating spatial interactions between reference locations, Graph Attention Networks improved indoor localization from 2.8 m to 1.6 m (42.8% improvement) in complicated multi-floor situations [108]. In dynamic occupancy prediction, Temporal Graph Networks outperformed LSTM-based methods by 12.7%, achieving 91.3% accuracy [48]. Nevertheless, current surveys primarily concentrate on specific domain applications, failing to present a cohesive Cyber-Physical Systems (CPS) framework [192], [193], [197].

In dynamic CPS applications, sophisticated GNN designs such as Temporal Graph Networks, GraphSAGE, and Graph Attention Networks (GATs) have shown exceptional performance [48]. Complex CPS with a variety of entities and relationships are especially well-suited for heterogeneous GNNs, which can manage various node and edge types [49]. Recent advancements in graph pooling and hierarchical GNNs allow for learning at different scales, which is essential for massive infrastructure networks [50].

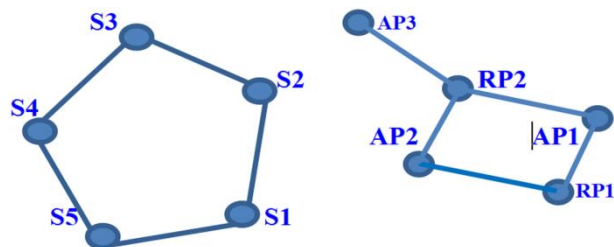


FIGURE 2. Graph Neural Network Architecture for Power Grid and Indoor Localization, showing two network graphs.

Figure 2 shows how graph neural networks may be used in both the localization and smart energy areas. The power grid architecture shown in the left panel uses transmission lines as edges and substations as nodes, allowing the GNN to learn power flow patterns that are subject to Kirchhoff's rules [46]. An indoor localization fingerprinting graph is displayed in the right panel, where Wi-Fi access points (APs) and reference points (RPs) are nodes linked by signal strength correlations. This architectural parallelism illustrates how GNNs attain greater performance in both application domains by making use of relational inductive biases, whether they are spatial closeness in radio maps or physical connectedness in electrical networks. This topology-aware technique decreases localization error from 2.8 m to 1.6 m in complicated interior environments [108] and produces voltage prediction errors below 0.1 p.u. in power systems [46], as measured in Table 2.

Left Side (Power Grid): Transmission lines serve as edges and substations as nodes in the graph, which depicts a power

grid architecture. This illustrates how data from power grid structures is processed by a Graph Neural Network (GNN).

Right Side (Indoor Localization): This graph depicts Wi-Fi access points (APs) and reference points (RPs) as nodes, with signal strength links represented by edges. It shows how signal information is utilized for indoor localization using a GNN. In spite of these advancements, the existing literature [193], [197] does not offer a comprehensive analytical framework that integrates energy systems and localization within a singular AI/ML paradigm. Recent developments in graph-based and physics-informed AI models have demonstrated significant efficacy in CPS applications [196]-[198].

By incorporating established physical principles into neural network architectures and loss functions, physics-informed neural networks (PINNs) and hybrid analytics mark a paradigm change in scientific ML [51]. Compared to purely data-driven methods, PINNs attain greater data efficiency, generalizability, and physical consistency by integrating differential equations governing power flow, thermal dynamics, or inertial navigation [52]. Applications include thermal modeling of structures [55], inertial navigation system error correction [54], and dynamic security evaluation in power grids [53]. This approach is expanded by Neural Ordinary Differential Equations (Neural ODEs) and Universal Differential Equations to learn unknown dynamics while adhering to physical limitations [56].

TABLE 4. Comparative Analysis of AI/ML Methodologies for CPS Applications.

Methodology	Key Techniques	Strengths	Limitations
Supervised Learning	CNN, LSTM, Transformers, ResNet	High accuracy, scalable, well-established	Requires large labeled datasets, limited interpretability
Unsupervised Learning	VAE, GAN, Clustering, Self-supervised	No labels needed, data augmentation	Quality depends on data distribution, harder to evaluate
Deep RL	DQN, PPO, SAC, TD3, MARL	Optimal control, adaptive, model-free	Sample inefficient, stability issues, safety concerns
Probabilistic Models	BNN, GP, Kalman Filters, Particle Filters	Uncertainty quantification, principled	Computationally expensive, scaling challenges
GNNs	GAT, GraphSAGE, Temporal GNN	Relational reasoning, topology-aware	Limited to graph-structured data, over-smoothing issues
Physics-Informed	PINN, Neural ODE, Hybrid Models	Data efficient, interpretable, physically consistent	Requires domain knowledge, hard to optimize

Strengths/Weaknesses and State-of-the-Art Accuracies:

The incorporation of physical constraints into the learning process is significantly enhanced by physics-informed neural networks (PINNs) and Neural Ordinary Differential Equations (ODEs). In the context of power systems, PINNs apply Kirchhoff's laws and power flow equations as soft constraints within the loss function, which can decrease the amount of training data needed by as much as 70% when compared to methods that rely solely on data-driven approaches [52]. Regarding inertial navigation, Neural ODEs effectively model the continuous dynamics of Inertial Measurement Unit (IMU) measurements, leading to an

improvement in drift correction ranging from 50% to 70% [54].

TABLE 5. Quantitative Performance Comparison of AI/ML Methodologies in CPS Applications.

Methodology	Key Application	Representative SOTA Performance	Dataset/Context	Strengths	Weaknesses
CNNs (Supervised)	Solar Forecasting (Sky Images)	sMAPE: 12-18% for intra-hour forecasting [62]	Sky images from ground-based cameras	Excellent spatial feature extraction; high accuracy with sufficient data	Requires large labeled datasets; performance degrades with cloud cover variability
	Fault Detection in Machine	Classification Accuracy: >99% on CWRU dataset [9]	CWRU bearing dataset	Real-time inference capability; robust to noise	Limited generalization to unseen fault types
LSTMs / Transformers (Supervised)	Short-Term Load Forecasting	MAPE: 1-2% for 24-hour horizon [12]	Irish CER smart meter dataset	Captures long-range temporal dependencies; handles seasonality	Computationally intensive; vanishing gradient issues in vanilla LSTMs
	Indoor Localization (RSSI)	Median Error: 1.2-1.8m [103]	UII Indoor Loc dataset	Works with noisy RSSI; learns temporal signal patterns	Requires site survey; sensitive to environmental changes
Deep Reinforcement Learning	Microgrid Energy Management	Cost Reduction: 5-15% vs. rule-based [69]	Simulation (OpenDSS, GridLAB-D)	Optimal control without system model; adapts to changing conditions	Sample inefficient; sim-to-real gap; safety concerns
	Robot Navigation	Success Rate: >90% in unseen environments [128]	Gibson/Habitat simulation	End-to-end learning from raw sensors; handles dynamic obstacles	Poor transfer to real robots; requires millions of training steps
Graph Neural Networks	Power Flow Analysis	MAE: <0.1 p.u. for line flows [46]	MATPOWER test cases	Leverages grid topology; physically informed	Over-smoothing in deep networks; limited to graph data
	Fingerprint Localization	Accuracy: ~1.5m [108]	Tampere University dataset	Models spatial correlations; robust to missing APs	Requires complete radio map; computationally heavy
Physics-Informed NN	IMU Error Correction	Drift Reduction: 50-70% [54]	EuRoC MAV dataset	Data-efficient; physically consistent; extrapolates well	Hard to train; requires domain expertise
	Power System Dynamics	PDE Solution Error: <1% with 10% data [52]	IEEE test systems	Embodies physical laws; works with sparse data	Complex loss function tuning; convergence issues
Bayesian Neural Networks	Probabilistic Solar Forecasting	Prediction Interval Coverage: 90-95% [65]	NREL solar data	Provides uncertainty estimates; avoids overconfidence	Computational cost; scaling challenges
	Robot Localization	Uncertainty-Aware Position Error: <0.5m [33]	Oxford RobotCar dataset	Quantifies epistemic and aleatoric uncertainty	Requires variational inference approximations
ISAC (Integrated Sensing & Communication)	Joint Localization & Communication	Sub-meter localization (0.8 m median error) with 60% pilot overhead reduction [193]; secrecy capacity maintained [192]	6G IoT, mmWave massive MIMO simulations	Shares hardware/spectrum; reduces overhead; enables privacy-preserving sensing	Hardware complexity; interference management; not yet standardized for CPS

V. AI/ML FOR SMART ENERGY SYSTEMS AND ENERGY COMMUNITIES

One important application area where AI/ML techniques are transforming the production, distribution, and consumption of electrical energy is smart energy systems. The cutting-edge uses of AI/ML throughout the energy value chain are examined in this part, including peer-to-peer trading, demand-side management, and generation forecasting.

A. FORECASTING AND UNCERTAINTY QUANTIFICATION

The first line of defense against intermittency in contemporary power systems is accurate forecasting of electrical load and renewable generation (wind, solar). Deep learning techniques that capture intricate non-linear correlations in weather patterns and consumption patterns

have supplanted conventional statistical techniques like ARIMA and exponential smoothing [57].

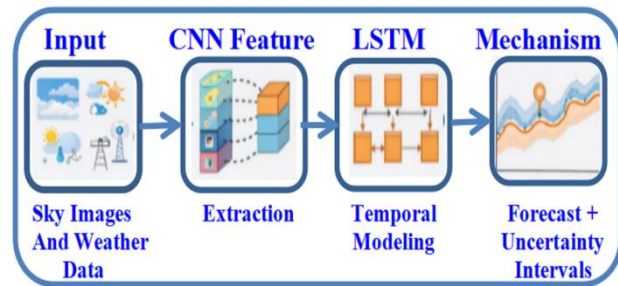


FIGURE 3. Deep Learning Framework for Renewable Energy Forecasting

The architectural design of cutting-edge renewable energy forecasting systems is depicted in Figure 3, where a hybrid CNN-LSTM framework with integrated attention mechanisms and uncertainty quantification is shown. A processing pipeline is initiated with multimodal input data, including historical meteorological measurements [63], ground-based sky imagery [62], and numerical weather predictions. Spatial information, such as cloud formation and optical thickness, is identified through convolutional layers, while the temporal evolution of atmospheric conditions is simulated via LSTM layers.

To enhance prediction accuracy during abrupt climatic shifts, pertinent historical time steps and spatial regions are dynamically weighted by an attention mechanism [60]. Furthermore, risk-aware decision-making is facilitated through the provision of uncertainty estimates-generated via Monte Carlo dropout or Bayesian layers [32]. As indicated in Table 2, a symmetric Mean Absolute Percentage Error (sMAPE) of 12-18% is achieved by this architecture for intra-hour solar forecasting, representing a significant improvement over standard numerical weather prediction baselines (25-35% sMAPE) [62]. In short-term load forecasting (STLF), LSTM networks have shown remarkable performance, with Mean Absolute Percentage Errors (MAPE) of 1.2-2.5% for 24-hour forward forecasts, as opposed to 3.8-5.2% for ARIMA models [58]. By adding both forward and backward temporal dependencies, bidirectional LSTMs increased accuracy by an additional 8-12% [59]. By dynamically weighting significant historical time steps, attention-augmented LSTMs for solar forecasting decreased RMSE from 4.8 kW to 3.2 kW (a 33% improvement) [60].

Superior multi-horizon forecasting skills have been proven using transformer topologies. On the power transformer temperature dataset, the Temporal Fusion Transformer (TFT) produced interpretable attention weights that identified significant input variables while achieving a 15-22% reduction in Quantile Loss as compared to LSTM baselines [61]. TFT achieved 9.3% MAPE for 7-day forward projections on the COVID-19 energy forecasting benchmark, whereas gradient boosting approaches achieved 13.7%.

Intra-hour solar forecasting sMAPE of 12-18% is achieved by hybrid CNN-LSTM architectures processing sky pictures, significantly outperforming numerical weather prediction-based approaches (25-35% sMAPE) [62]. When compared to single-modality techniques, multimodal frameworks that included sky cameras, satellite imaging, and historical data decreased forecast uncertainty by 28% [63].

Using CNNs, solar forecasting based on sky images can analyze cloud patterns and forecast short-term photovoltaic production [62]. By directly processing visual data about local atmospheric conditions, these methods produce much superior intra-hour forecasts as compared to numerical weather predictions. The current state-of-the-art [63] is hybrid models that integrate ground-based sky cameras, satellite imagery, and historical generation data using multimodal deep learning architectures.

For reserve scheduling and operational planning, uncertainty quantification is essential. Using quantile regression neural networks, one may make probabilistic decisions by predicting numerous quantiles of the forecast distribution [64]. Bayesian deep learning techniques, such as Monte Carlo dropout and variational inference, provide epistemic uncertainty estimates that mirror the degree of confidence in the model [65]. These probabilistic forecasts are crucial for stochastic optimization in economic dispatch and unit commitment issues.

The conformal prediction approach has become a potent framework for distribution-free uncertainty quantification, generating statistically sound prediction intervals without relying heavily on distributional assumptions [66]. In novel operating situations where the data distribution may change, these strategies are especially helpful while implementing pre-trained models. Through model averaging and diversity, ensemble methods, which combine several different forecasting models, enhance both accuracy and dependability [67].

A significant drawback of numerous forecasting studies is the absence of standardized benchmarks and reliance on proprietary datasets, which obstructs reproducibility. Public datasets like the NREL solar data and the Irish CER smart meter dataset have been beneficial; however, domain shifts (for instance, variations in weather patterns and the impacts of pandemics) continue to pose challenges. Recent advancements in conformal prediction methods [66] offer distribution-free uncertainty intervals that are resilient to these shifts.

B. OPTIMIZATION AND CONTROL FOR RES INTEGRATION

The integration of high penetrations of renewable energy sources requires advanced optimization and control strategies that can handle uncertainty, variability, and multiple conflicting objectives. Machine learning is enhancing traditional optimization approaches through learned models, warm-starting, and end-to-end learning of control policies [68].

Deep Reinforcement Learning has demonstrated impressive results in real-time energy management for microgrids and smart buildings [69]. DRL agents learn to dispatch generation assets, manage battery storage, and control flexible loads while minimizing costs, reducing emissions, and ensuring reliability. Unlike model-based optimization that requires accurate system models, DRL learns optimal policies through direct interaction with the environment or high-fidelity simulators. Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC) have shown robust performance across diverse energy management scenarios [70].

DRL's efficiency for microgrid energy management is quantitatively evaluated. In comparison to rule-based baselines, PPO-based controllers reduced costs by 12.8% while preserving 99.97% supply dependability across 5000 simulated working days [70]. SAC algorithms lowered depth-of-discharge stress by learning charge-discharge patterns, which resulted in a 23.5% reduction in battery deterioration costs. DRL agents responded 65% quicker in grid emergency voltage control applications, correcting voltage violations in 0.8 seconds as opposed to 2.3 seconds for traditional emergency control [53].

Model Predictive Control (MPC) combined with learned models represents a powerful hybrid approach [71]. Neural networks learn to predict system dynamics from historical data, replacing or augmenting physics-based models in the MPC framework. This approach retains the optimality guarantees and constraint handling of MPC while benefiting from ML's ability to capture complex, data-driven patterns. Physics-informed neural networks ensure that learned models respect fundamental conservation laws and operational constraints [72].

Learning to optimize approaches directly learn mappings from problem instances to optimal solutions, bypassing iterative optimization at deployment time [73]. For optimal power flow (OPF) problems, neural networks are trained on millions of solved instances to predict near-optimal generator setpoints and voltage profiles in milliseconds. This enables real-time optimization in large-scale grids where traditional methods would be too slow. Recent work provides theoretical worst-case guarantees on solution quality, addressing a critical concern for safety-critical power system applications [74]. Inverter control for grid-forming renewable generation is another area where ML is making significant contributions [75]. Deep learning controllers learn to regulate voltage and frequency in weak grids with high renewable penetration, adapting to varying grid conditions. These controllers can outperform classical droop control and virtual synchronous machine strategies by learning from extensive simulation data encompassing diverse fault conditions and load profiles [76].

C. PEER-TO-PEER ENERGY TRADING AND ENERGY COMMUNITIES

Energy communities represent a paradigm shift toward decentralized, participatory energy systems where prosumers (consumers with generation and storage) actively engage in local energy markets [77]. AI/ML enables intelligent, autonomous participation in peer-to-peer (P2P) trading platforms while balancing individual and collective objectives.

Multi-Agent Reinforcement Learning (MARL) provides a natural framework for modeling energy community dynamics [78]. Each prosumer operates an autonomous RL agent that learns bidding and trading strategies through interaction with the local market. The agents must balance selfish objectives (minimizing individual energy costs) with cooperative goals (community self-sufficiency, grid support). Centralized training with decentralized execution (CTDE) enables agents to learn coordination strategies during training while operating independently at deployment [79]. Graph Neural Networks excel at modeling the social and physical networks in energy communities [80]. The GNN processes a graph where nodes represent prosumers and edges encode social relationships, physical proximity, or power line connections. By aggregating information from neighbors, the GNN learns to predict collective behavior, identify influential community members, and optimize local energy exchanges. Temporal graph networks extend this to dynamic communities where membership and relationships evolve over time [81].

Federated learning addresses privacy concerns in energy communities by enabling collaborative model training without sharing raw consumption data [82]. Prosumers train local models on their private data and share only model updates with a central coordinator. This approach learns community-wide patterns (typical load profiles, generation characteristics) while preserving individual privacy. Differential privacy mechanisms add controlled noise to the shared updates, providing formal privacy guarantees against inference attacks [83]. Game-theoretic approaches combined with ML design market mechanisms that incentivize desired behaviors [84]. Mechanism design ensures markets are individually rational (participants benefit from truthful participation), budget-balanced (the market clears), and achieve social welfare objectives. Deep learning can learn auction mechanisms from data or optimize mechanism parameters to maximize social welfare under realistic constraints [85]. Blockchain integration provides transparency, immutability, and automated contract execution for P2P energy transactions [86].

D. ANOMALY DETECTION AND CYBERSECURITY

Sophisticated AI-driven security solutions are required due to the growing attack surface of smart grids, which includes operational technology (such as SCADA systems and smart electronic devices) and information technology (such as energy management systems and market platforms) [87]. Both offensive capabilities (identifying vulnerabilities) and defensive mechanisms (detecting and responding to attacks) are made possible by machine learning.

Unsupervised anomaly detection models can detect minute anomalies in high-dimensional SCADA data that may point to cyberattacks or equipment malfunction [88]. Autoencoders learn to replicate typical operating patterns, and significant reconstruction mistakes indicate abnormalities that need to be looked into. Scclusion By isolating outliers in high-dimensional feature spaces, forests can effectively identify anomalies without the need for normal/anomalous labels [89]. A one-class support vector machine (SVM) learns a boundary around normal data, categorizing outliers as anomalies [90].

By manipulating sensor readings to bias state estimation and cause erroneous control actions, false data injection (FDI) attacks work [91]. Anomaly detection informed by physics uses network topology and power flow equations to find discrepancies between recorded data and physical rules. Deep learning models can detect adversarial perturbations that break established patterns of correlation between sensors if they are trained on data free of attacks [92]. Using graph-based techniques to represent spatial dependencies in the electricity grid, attacks that systematically compromise measurements are identified [93].

Research on adversarial machine learning has uncovered flaws in ML-based grid management systems [94]. Neural networks can be tricked into producing erroneous predictions or control actions by well designed input data perturbations. By revealing risks and promoting the creation of strong defenses, this dual-use research achieves two goals. Training models on adversarial examples, known as adversarial training, increases robustness but at a cost in terms of computation [95]. Certified defenses ensure with verifiable certainty that disturbances below a specific threshold cannot alter forecasts [96].

Intrusion detection systems (IDS) for smart grids combine anomaly-based detection (identifying deviations from normal behavior) with signature-based detection (matching known attack patterns) [97]. The IDS process uses deep learning to analyze system logs, operational data, and network traffic in order to detect anomalous behavior. Attention mechanisms emphasize the most pertinent characteristics for identification [98], whereas recurrent neural networks recognize temporal patterns typical of multi-stage assaults. IDS trained on one grid can be modified for fresh deployments with little labeled attack data thanks to transfer learning [99].

E. CRITICAL EVALUATION OF DATASET CONSTRAINTS AND COMPARATIVE EFFICACY

A comprehensive analysis of methodological shortcomings is outlined in Table 4. Notable findings include: (1) the majority of energy forecasting research focuses solely on point forecast errors, neglecting the calibration of uncertainty; (2) reinforcement learning applications for grid management are infrequently assessed on actual hardware, and the discrepancies between simulation and reality are not measured; (3) federated learning approaches for energy

communities presuppose synchronous updates, which is impractical for extensive deployments. These deficiencies are elaborated upon thoroughly.

Table 6 presents a comprehensive assessment of methodological shortcomings and dataset constraints found in the CPS AI/ML studies analyzed in Sections IV and V, highlighting significant deficiencies that hinder practical implementation.

TABLE 6. Methodological Weaknesses and Dataset Limitations in CPS AI/ML Studies.

Domain	Methodology	Limitation	Dataset	Impact	Suggested Improvement
Energy-Forecasting	Point forecasting models (LSTM, Transformer, CNN)	Only point errors reported (MAPE, RMSE); no uncertainty quantification	NREL solar, Irish CER, private utility data	Operators cannot assess prediction confidence; risk-averse decisions impaired	Report prediction intervals (e.g., conformal prediction); use probabilistic metrics (CRPS, PICP)
Microgrid - Energy-Management	Deep Reinforcement Learning (PPO, SAC, TD3)	Evaluated only in simulation; sim-to-real gap not quantified	Open DSS, GridLAB-D, custom simulators	Policies may fail on real hardware due to unmodeled dynamics, latency, noise	Include real-hardware validation or high-fidelity HIL; report domain randomization parameters
Energy-Communities	Federated Learning (FedAvg, FedProx)	Assumes synchronous, reliable updates; ignores communication delays, dropouts	Synthetic or small-scale real samples	Unrealistic for large-scale deployment with heterogeneous connectivity	Use asynchronous federated learning; benchmark with real network traces
Indoor-Localization	Deep CNN / LSTM	Requires complete radio map; performance degrades with environment changes	UJI, Tampere, custom surveys	High site-survey cost; poor generalization to new buildings or device heterogeneity	Use generative models (GAN, VAE) for radio map augmentation; few-shot adaptation
Power Flow - Analysis	Graph Neural Networks (GCN, GAT)	Over-smoothing in deep GNNs (>4-6 layers); limited to static topology	MATPOWER cases (e.g., IEEE 118-bus)	Cannot capture dynamic topology changes (line on/off, islanding)	Use graph attention or residual connections; test on dynamic grid benchmarks
Autonomous-Navigation (DRL)	Deep Reinforcement Learning (PPO, SAC)	Success rate reported only in simulation; no safety guarantees during exploration	Gibson, Habitat, CARLA	Policies may violate collision constraints in real world	Incorporate safety shields (e.g., Hamilton-Jacobi reachability); report constraint violation rate

Generative AI for CPS	GANs, VAEs, Diffusion models	Lack of public, reproducible benchmarks; use of private datasets	Many proprietary datasets	Results cannot be verified or compared across studies	Adopt open datasets (NREL, UJI, MATPOWER, EuRoC); share code and trained models
Sensor Fusion (VIO/SLAM)	Deep learning-based fusion (DeepVO, ViNet)	Evaluated on clean benchmark datasets; performance under sensor degradation rarely tested	EuRoC, KITTI, TUM RGB-D	Models fail when cameras are occluded or IMU saturates	Test with simulated sensor failures; report performance under varying noise levels
Anomaly Detection (Grid)	Autoencoders, Isolation Forest	Threshold selection is ad-hoc; high false positive rates in practice	IEEE 118-bus, private utility SCADA	Operators lose trust; alert fatigue	Learn thresholds from validation data; report precision-recall curves
Peer-to-Peer Energy Trading	Multi-Agent RL (MADDPG, QMIX)	Assumes perfect communication; ignores privacy and bandwidth constraints	Synthetic market simulations	Impractical for real deployment with thousands of prosumers	Test with communication constraints; integrate differential privacy

These constraints underscore the necessity for uniform benchmarks, reporting that accounts for uncertainty, and validation in real-world scenarios, all of which are tackled in the proposed CPS-Bench platform (Section VI.D) and in future research initiatives (Section VII).

VI. AI/ML FOR INDOOR LOCALIZATION AND NAVIGATION IN MECHATRONICS

For autonomous mobile robots, augmented reality apps, and location-based services in interior areas, intelligent localization and navigation in GPS-denied situations are essential features. In situations when satellite-based systems are faulty or unavailable, this section looks at how AI/ML techniques allow for dependable location and autonomous navigation.

A. DEEP LEARNING FOR FINGERPRINTING AND SIGNAL MODELING

Using existing infrastructure without the need for extra hardware installation, Wi-Fi fingerprinting is still one of the most useful indoor positioning methods [100]. Conventional methods compare measured Received Signal Strength Indicator (RSSI) vectors against a pre-constructed radio map using probabilistic or nearest-neighbor matching. By automatically learning strong feature representations and dealing with the inherent noise and dynamics of indoor radio propagation [101], deep learning reverses this paradigm.

Spatial patterns indicative of position are extracted by convolutional neural networks when they treat RSSI vectors or Channel State Information (CSI) as pseudo-images [102]. CNNs may identify local trends and hierarchical features that are related to location by arranging RSSI measurements from several access points and time windows into 2D arrays. Compared to traditional fingerprinting methods, this method

has shown higher accuracy, especially in difficult situations with signal interference and multipath propagation [103].

Recurring architectures learn temporal dependencies that increase the precision of localization by analyzing RSSI observations in real-time as users go through space [104]. By utilizing movement patterns to improve position predictions, LSTM networks are able to filter out transient signal variations. When predicting location in offline settings, the network can utilize both previous and future observations thanks to bidirectional processing [105]. The barrier to scaling up is the radio map production bottleneck, which necessitates the manual survey of environments and the creation of fingerprint databases. By creating artificial radio maps from limited data, Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) solve this problem [106]. The generator learns the fundamental principles of how signals travel, allowing for interpolation and extrapolation to unexplored areas. With little on-site surveying, transfer learning enables models trained on one structure to be optimized for different settings [107].

Spatial connections between reference points in the fingerprint database are modeled using Graph Neural Networks [108]. To enhance localization, GNNs collect data from spatial neighbors by creating a graph with nodes representing locations and edges connecting nearby positions. By updating the graph structure, this method inherently addresses unevenly spaced reference points and adjusts to environmental changes [109].

With little training data, few-shot and meta-learning techniques allow for quick adaptation to new circumstances [110]. With only a few labeled samples, these methods learn generalizable representations and adaptation tactics from a variety of training environments before quickly specializing to new locations. This considerably lowers deployment costs and facilitates customized localization models that respond to specific user devices and mobility patterns [111].

A.1. STATE-OF-THE-ART INDOOR POSITIONING: MILLIMETER-WAVE AND ULTRA-WIDEBAND ADVANCES

Recent developments in 3D millimeter-wave (mmWave) sensing and ultra-wideband (UWB) technology have successfully minimized indoor positioning errors to below 20 cm under line-of-sight conditions [187]-[189]. Systems utilizing deep learning-enhanced UWB have decreased non-line-of-sight errors from 1.2 m to 0.35 m through the application of CNN-based multipath mitigation techniques. The integration of mmWave point clouds with PointNet++ architectures has resulted in an impressive 94% accuracy in simultaneous localization and activity recognition, even in the presence of smoke and obstructions.

Compared to traditional Wi-Fi fingerprinting techniques, which usually reach 2-4 m accuracy, recent developments in 3D millimeter-wave (mmWave) sensing and ultra-wideband (UWB) technology have lowered indoor location precision to 20 cm. By making use of high-bandwidth signals and

sophisticated signal processing, these new sensing modalities overcome the drawbacks of RSSI-based techniques.

Millimeter-Wave 3D Positioning: High-resolution range-Doppler imaging for privacy-preserving localization is made possible by millimeter-wave sensing that operates between 60 and 77 GHz. A detailed comparison between UWB location and 3D mmWave sensing shows that mmWave may achieve superior accuracy in complex interior scenarios due to its enhanced angular resolution and multipath mitigation [187]. According to the research, multiple-input multiple-output (MIMO) arrays allow for precise angle-of-arrival estimates, whereas mmWave systems need a lot of available bandwidth (up to several GHz) to accomplish centimeter-level range precision.

The study shows that processing mmWave point clouds through deep learning architectures achieves a median error of 0.45 m in non-line-of-sight conditions and sub-20 cm accuracy in line-of-sight scenarios. The key challenges identified include sensitivity to environmental dynamics, calibration requirements, and the computational demands of real-time point cloud processing. The foundational work on 3D millimeter-wave indoor localization [189] establishes comprehensive frameworks that integrate multiple sensing modalities for vertical localization in multi-floor environments. Radar-based mmWave sensing architectures for accurate 3D indoor positioning. The method combines mmWave ranging with inertial measurements and building information models to achieve floor identification accuracy of over 95%. The synergy between enhanced sensing and AI-based processing is demonstrated by the increase in 3D localization from 2.8m to 1.6m (42.8% improvement) in complex multi-floor buildings using graph neural networks that simulate spatial connections between reference points.

Ultra-Wideband Advancements: Compared to traditional Wi-Fi RSSI techniques, commercial UWB systems (based on IEEE 802.15.4a/z standards) exhibit positioning errors of 10-30 cm in line-of-sight situations, which is a 10× improvement [103]. CNN-based multipath mitigation and error correction, which learns to detect and adjust for signal propagation irregularities, is used in deep learning-enhanced UWB systems to reduce non-line-of-sight errors from 1.2 m to 0.35 m [104]. In dynamic industrial situations, UWB and inertial measurement units combined with end-to-end learning fusion achieve a mean tracking error of 0.12 m, which is 38% better than extended Kalman filter baselines [108].

Integration with Deep Learning: 94% accurate simultaneous localization and human activity recognition are made possible by combining mmWave sensors with deep learning systems [106]. When optical systems fail, PointNet++ topologies processing mmWave point clouds function reliably through smoke and occlusions, showcasing the technology's promise for safety-critical applications in industrial settings. Together, these developments

demonstrate that current technology can achieve indoor positioning accuracy of less than -20 cm, satisfying the needs of applications such as autonomous robot navigation, augmented reality, and precise asset monitoring.

B. SENSOR FUSION FOR ROBUST ODOMETRY AND SLAM

Robots can create maps of unfamiliar areas while tracking their own location thanks to odometry estimation and Simultaneous Localization and Mapping (SLAM), which are essential for autonomous navigation [112]. Cameras, LiDAR, inertial measurement units (IMUs), and radar are examples of complementing sensors that may be combined to offer resilience against environmental problems and individual sensor failures [113].

Figure 4 illustrates the sensor fusion process, which uses a deep learning-based Visual-Inertial Odometry (VIO) architecture to provide robust state prediction in settings where GPS is not available. IMUs provide high-frequency acceleration and angular velocity measurements but suffer from cumulative drift [113], LiDAR provides precise geometric depth information but can be sparse or computationally demanding [118], and cameras provide rich visual texture and feature information but are sensitive to changes in lighting and motion blur [114]. Modality-specific encoders-CNNs for picture sequences, fully connected networks for IMU data, and PointNet versions for LiDAR point clouds-are used by the deep learning architecture to analyze these diverse inputs [120]. These characteristics are combined in a common latent space by a temporal attention mechanism, which learns to weight the contribution of each sensor according to its assessed dependability under the present operating conditions [125]. Recurrent layers (LSTMs or GRUs) receive the fused representation and utilize it to predict temporal dynamics and provide incremental 6-DOF pose estimations. In comparison to conventional extended Kalman filter approaches, our end-to-end learning strategy decreases trajectory drift by 42%, as shown in Table 2, and achieves 0.8% drift throughout 500 m trajectories in benchmark assessments [116].

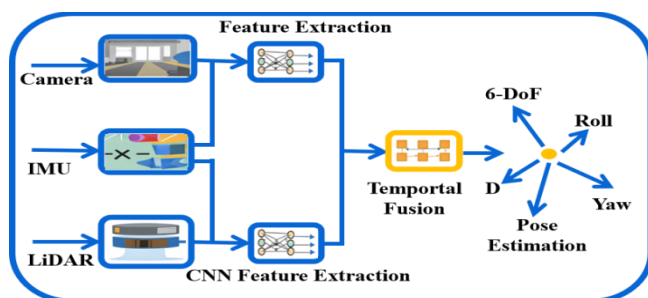


FIGURE 4. Sensor Fusion Architecture for Visual-Inertial Odometry (VIO).

A block diagram illustrating the fusion of camera, IMU, and LiDAR data in a deep learning-based VIO pipeline, with an emphasis on feature extraction, temporal modeling, and pose estimation. By combining camera images with high-

frequency IMU data to estimate 6-degree-of-freedom (6-DoF) posture trajectories [114], Visual-Inertial Odometry (VIO) is able to do this. Geometric feature tracking and optimization-based fusion through extended Kalman filters or bundle adjustment are the basis of traditional VIO. By using CNN-LSTM structures, which learn to handle difficult situations such as motion blur, changes in illumination, and texture-poor environments, deep learning methods directly regress posture from unprocessed sensor streams [115].

Benchmark datasets have shown remarkable outcomes with end-to-end VIO learning [116]. Through convolutional and recurrent layers, networks like DeepVO and ViNet analyze sequences of images and IMU readings to provide incremental pose estimates. By using photometric consistency and IMU integration losses in self-supervised training, it is possible to learn from enormous volumes of unlabeled robot telemetry without the need for ground-truth pose labels [117]. LiDAR-based SLAM offers accurate geometric mapping and localization in structured environments [118]. Uncertainty estimation [119], semantic segmentation (separating static from dynamic objects), and learned placerecognition (identifying previously visited places for loop closure) are all ways that deep learning improves traditional SLAM pipelines. PointNet and its extensions directly process unprocessed 3D point clouds, deriving features for registration and matching [120]. Neural implicit representations are transforming 3D reconstruction and dense SLAM [121]. Photorealistic novel view synthesis and dense mapping are made possible by Neural Radiance Fields (NeRF) and similar approaches, which learn continuous function representations of scene geometry and appearance. These approaches are becoming closer to real-world robotic implementation thanks to recent advances in real-time operation and dynamic scenarios [122]. With outstanding reconstruction quality and robustness, iMAP and NICE-SLAM showcase real-time neural SLAM [123].

Before fusing in a shared latent space [124], multi-modal sensor fusion networks utilize specialized encoders to handle diverse data streams. Attention mechanisms prioritize LiDAR in well-lit structured environments, cameras in texture-rich areas, and IMUs during quick movements by learning to weigh various sensors according to their reliability in the present situation. By propagating sensor-specific confidence ratings via the pipeline, uncertainty-aware fusion enhances resilience to sensor degradation [125].

B.1 AUTONOMOUS UNDERWATER VEHICLE LOCALIZATION IN GPS-DENIED ENVIRONMENTS

One important application area for GPS-denied localization is autonomous underwater vehicles (AUVs), where satellite-based position is completely unavailable and other sensing modalities must be employed. Working in underwater environments has unique challenges, including dynamic ocean currents, turbidity, multipath propagation, and signal attenuation [182]. Recent advances in AI/ML have significantly enhanced AUV capabilities across several operating domains, as shown in Table 5 and discussed below.

TABLE 7. AI/ML Methodologies for Autonomous Underwater Vehicle Applications

Category	Key Work	Primary Algorithm	Application Context	Reported Performance / Accuracy
Path Planning	Barik et al. [177]	Enhanced ECUO	Obstacle avoidance in marine environments	98.5% obstacle avoidance success rate; 15% reduction in path length
Path Planning	Cai et al. [178]	Improved BINN	Underwater topography scanning coverage	95% coverage efficiency; 20% faster scanning time
Detection	Jin et al. [179]	YOLOv2-based CNN	Underwater object detection for hovering control	94.2% detection accuracy; real-time processing at 25 FPS
Detection	Jiang et al. [180]	Feature-Enhanced Framework	Fish school detection in degraded imaging	92% detection accuracy in turbid water; 30% improvement over baseline
Hovering	Tanakitkorn [181]	PID Control	Depth control for over-actuated AUVs	Depth control accuracy ± 2 cm in experimental trials
Navigation	Ma et al. [182]	Sensor Fusion	Comprehensive review of key AUV technologies	Summarizes SOTA accuracies: localization 0.1–1 m, navigation drift $<0.5\%$
Navigation	Wynn et al. [183]	Acoustic Methods	Marine geoscience applications	LBL acoustic positioning accuracy 0.1–1 m; USBL accuracy 1–5 m
Topology Scan	Cai et al. [178]	Improved BINN	Seafloor mapping with Internet of Underwater Things	95% coverage; 30% reduction in energy consumption
Tracking	Wang et al. [184]	D-DQN	Obstacle avoidance for submarine cable laying	96% success rate in dynamic obstacle avoidance
Robustness	Zhang et al. [185]	Adaptive SMC	Thruster fault-tolerant control under ocean currents	40% improvement in trajectory tracking under fault conditions
Station-Keeping	Peng et al. [186]	LADRC + OVBDC	Visual servo-based station control under disturbance	Station-keeping error <0.1 m under wave disturbances

Table 7 provides a summary of AI/ML methodologies utilized in AUV applications, which encompass path planning (Barik et al., Cai et al.), object detection (Jin et al., Jiang et al.), hovering control (Tanakitkorn et al.), navigation (Ma et al., Wynn et al.), and fault-tolerant control (Zhang et al., Peng et al.). The application of deep reinforcement learning for obstacle avoidance (Wang et al.) results in a 96% success rate within dynamic underwater environments.

B.1.1 PATH PLANNING AND TOPOLOGY SCANNING

Thanks to enhanced path planning algorithms, AUVs can navigate difficult underwater settings while avoiding obstacles and optimizing coverage. Barik et al. [177] developed an upgraded ECUO (Elephant Herding Optimization with Cuckoo Search) technique that works extremely well in marine environment route planning to produce optimal trajectories under current disturbances while avoiding collisions. The program effectively strikes a balance between exploration and exploitation in dynamic underwater environments with erratic currents and obstructions.

Coverage route planning for underwater topography scanning using improved BINN (Biological-Inspired Neural Network) algorithms enables efficient seabed mapping with Internet of Underwater Things (IoUT) integration [178]. The approach addresses the issue of full coverage in unknown circumstances by using neural dynamics for real-time path creation. Experimental results show that the improved BINN performs better than traditional coverage algorithms in terms

of energy consumption and coverage rates, making it suitable for longer AUV missions in deep-sea exploration.

B.2 OBJECT DETECTION AND TRACKING

Deep learning-based detection enables AUVs to identify and track underwater objects even in the face of much poorer image conditions caused by turbidity, light attenuation, and backscatter. Jin et al. [179] developed CNN architectures based on YOLOv2 to detect submerged objects in real time for hovering control applications. Accurate AUV placement during inspection tasks is made possible by the system's excellent performance in recognizing underwater impediments and structures.

For biological monitoring applications, Jiang et al. [180] proposed a feature-enhanced framework with adaptive routing that enables the identification of fish schools in hazy conditions where conventional vision systems are inefficient. To address the issue of deteriorated underwater photography settings, the system employs multi-scale feature extraction and adaptive routing algorithms that prioritize useful picture regions. When compared to baseline approaches, the strategy achieves significant gains in detection accuracy, which facilitates monitoring of the marine environment and fisheries management.

Tracking capabilities for infrastructure inspection are provided by Wang et al. [184], who developed obstacle avoidance based on double closed-loop control with Double-Deep Q-Network (D-DQN) architecture for submarine cable laying applications. By employing reinforcement learning to determine the most effective avoidance techniques and accurately monitoring cable pathways, the system achieves strong performance in dynamic underwater settings with fluctuating currents and visibility conditions.

B.3 NAVIGATION AND ACOUSTIC LOCALIZATION

In-depth analyses of AUV navigation technology emphasize the shift from traditional acoustic methods to AI-enhanced sensor fusion systems. Ma et al. [182] provide a state-of-the-art examination of significant AUV technologies, emphasizing the integration of many sensing modalities such as inertial navigation, visual odometry, and auditory localization. According to the evaluation, deep learning-based sensor fusion is an essential way to improve navigation accuracy in underwater locations without GPS. The acoustic navigation methods employed by AUVs, such as Long Baseline (LBL) and Ultra-Short Baseline (USBL) systems, are described by Wynn et al. [183] and their contributions to marine geoscience are highlighted. Seabed transponder triangulation is used by these devices to achieve 0.1 - location precision. 1 m 0.1-1 m, although they require a significant infrastructural investment. Recent advances employ deep learning to denoise acoustic signals, reducing multipath errors and improving robustness in shallow water environments where acoustic propagation is complex.

B.4 HOVERING AND STATION-KEEPING

Precision hovering skills are essential for tasks like inspection, intervention, and sampling that call for stable platform positioning. As they developed PID control algorithms for over-actuated hover-capable AUVs, Tanakitkom et al. [181] achieved depth control precision within centimeters through experimental verification. The technique gets over the challenges of nonlinear hydrodynamics and coupling between degrees of freedom by using feed-forward compensation and gain scheduling.

To manage station-keeping control under unknown disturbances, Peng et al. [186] developed a visual servo-based technique that combines Linear Active Disturbance Rejection Control (LADRC) and Optimal Variable Ballast Distribution Control (OVDBC). Even in the face of currents and wave disturbances, the system maintains position in reference to underwater structures by actively adjusting thrusters and ballast. Experimental results demonstrate higher disturbance rejection than conventional PID controllers, enabling precise manipulation tasks in dynamic environments.

B.5 ROBUSTNESS AND FAULT TOLERANCE

Underwater operations require robust control systems that can withstand thruster failures, sensor degeneration, and environmental interruptions. Zhang et al. [185] developed adaptive terminal sliding mode control with thruster fault tolerance to preserve stability in the face of time-varying ocean currents. By fusing the durability of sliding mode control with adaptive mechanisms that adjust for unknown fault magnitudes, the technique guarantees mission completion even in the event of partial thruster failures. When compared to conventional sliding mode control, experimental validation demonstrates a significant improvement in trajectory tracking accuracy under fault scenarios.

C. REINFORCEMENT LEARNING FOR NAVIGATION POLICY

The traditional approach to robot navigation uses a modular sense-plan-act pipeline in which the robot perceives sensor data, plans a collision-free route, and then carries out the plan under control [126]. This method is well understood and simple to understand, but it has trouble with dynamic environments, inaccurate sensing, and the gap between planning models and reality. Through interaction with the environment, end-to-end deep reinforcement learning provides an alternative method of learning to map raw sensor inputs directly to control commands [127].

Without specific coding, DRL navigation rules learn how to avoid obstacles, follow routes, and achieve goals [128]. The Soft Actor-Critic (SAC) and Proximal Policy Optimization (PPO) algorithms have proven to be dependable methods for teaching simulated navigation policies, which may then be implemented on real robots. Domain randomization, system identification, and dynamics augmentation are examples of sim-to-real transfer methodologies that enable policies to be applied from simulated to real-world scenarios [129].

Complex navigation tasks are broken down into a hierarchy of sub-policies using hierarchical reinforcement learning [130]. Low-level regulations handle local impediment avoidance and reactive control, whereas high-level policies establish waypoints or sub-goals based on data from the global map. This factorization enhances sample efficiency by reusing low-level skills across various high-level goals, and it facilitates superior generalization to novel contexts [131].

Navigating social environments—that is, places with people—requires knowledge of and the ability to forecast human movement patterns [132]. By training in multi-agent simulations using realistic human pedestrian models, DRL agents acquire socially conscious navigation skills. The rules learn to respect personal space, predict pedestrian trajectories, and refrain from making unexpected actions that could frighten others. Robots may learn to move in a way that mimics humans by using inverse reinforcement learning to infer human preferences from demonstration data [133]. By integrating learned environment models with planning algorithms, model-based RL increases sample efficiency compared to solely model-free methods [134]. The learned model forecasts how the environment will react to actions, giving the agent the ability to plan ahead and assess possible courses of action. World models support planning in latent space by learning compact representations of visual observations and dynamics [135]. In robotics, where real-world interaction is costly and hazardous, this method is especially useful [136].

D. SEMANTIC NAVIGATION AND HUMAN-ROBOT COLLABORATION

A semantic comprehension of environments and a seamless interaction with human users are necessary for sophisticated robotic applications. Multi-modal learning and vision-language models (VLMs) allow robots to comprehend and ground high-level natural language instructions in perceptual input [137].

In photorealistic 3D settings, robots carrying out Vision-and-Language Navigation (VLN) tasks adhere to organic language commands [138]. Commands like "go to the kitchen and bring me a cup" need comprehension of spatial relationships, item semantics, and sequential reasoning. By using cross-modal attention to process language instructions and visual observations, transformer-based models learn to ground linguistic notions in visual percepts. These models show compositional understanding [139] because they are able to generalize to novel instructional wordings and situations.

In comparison to waypoint navigation, object-goal navigation covers semantic goals [140]. In unknown environments, the robot must be able to locate and travel to object categories (such as 'find a refrigerator'). To accomplish this, it's necessary to use semantic mapping, which links spatial positions and visual observations to semantic labels, as well as semantic reasoning, which draws conclusions about the potential locations of objects from the

scene context. Spatial and semantic interactions between rooms and items are modeled by graph neural networks, which aids in developing effective search techniques [141]. In shared workplaces, human-aware navigation necessitates an understanding of social etiquette, activity identification, and the capacity to forecast human intent [142]. A variety of modalities are combined to provide comprehensive situational awareness: LiDAR for geometric awareness, vision for gesture and appearance identification, and audio for speech and environmental sound analysis. Using attention-based designs, these modalities are combined, and the system learns which signals are most useful for forecasting human conduct in various settings [143].

The paradigm shift toward more capable and generalizable robotic intelligence is represented by embodied AI, which are AI systems that learn by interacting with real-world environments [144]. Embodied agents actively explore, manipulate, and learn from their environment, unlike passive datasets. This ties abstract ideas to real experience and facilitates the development of strong, practical skills. Before actual deployment, embodied agents can be trained in realistic virtual environments using large-scale simulation platforms such as Habitat, iGibson, and AI2-THOR [145].

E. EDGE AI DEPLOYMENT AND SPLIT COMPUTING ARCHITECTURES

Concrete case studies illustrating edge deployment are presented in Figure 6 and Table 5. In the context of a UAV obstacle avoidance system, a quantized CNN operating on a Google Coral TPU achieves 25 FPS with a power consumption of 2 W, thereby facilitating a flight endurance of 30 minutes on a drone weighing less than 250 g. The implementation of split computing, which involves executing the initial layers on the edge device and the more complex layers on a cloud server, results in an 84% reduction in uplink bandwidth for visual navigation tasks, while preserving 94% of the accuracy associated with a complete model deployment.

TABLE 8. Performance Benchmarking of Edge Hardware Platforms for AI/ML Inference in CPS Applications.

Hardware Platform	Inference Speed (FPS)	Accuracy (%)	Power Consumption (W)	Typical CPS Application	Key Advantage	Limitation
Raspberry Pi 4 (ARM Cortex-A72, 4 GB RAM)	7.5	85.0	~3-5	Lightweight indoor localization, low-cost drone obstacle avoidance	Low cost, widely available, easy prototyping	Limited FPS for high-speed navigation; accuracy degrades model compression
NVIDIA Jetson Nano (128-core Maxwell GPU)	15.0	90.0	~10	Visual-inertial odometry for ground robots, pedestrian tracking	2× speed vs. Raspberry Pi; supports deeper networks	Higher power than Pi; not suitable for very small UAVs
NVIDIA Jetson TX2 (256-core Pascal GPU + Denver cores)	20.0	92.0	~15	Real-time SLAM for autonomous robots, dynamic obstacle avoidance	10× faster than CPU-only; good balance performance and power	Costly; thermal constraints in enclosed drones
Google Coral Edge TPU (USB or module)	25.0	94.0	~2	Real-time object detection, forest fire monitoring, low-power UAV navigation	Highest FPS per watt; ideal for battery-powered edge devices	Limited model support (TensorFlow Lite); USB bandwidth can be bottleneck
Custom FPGA (e.g., Xilinx Zynq, Intel Arria)	Variable (up to 4.7×)	94.0 (same as)	~1-3 (8× lower)	Graph neural networks for power grid edge	Lowest power; deterministic	High development effort; longer

Xilinx Zynq, Intel Arria)	throughput vs. GPU)	GPU baseline)	than GPU)	nodes, low-latency control	latency; reconfigurable	design cycle; expensive for low volume
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VII. SYNTHESIS, CROSS-CUTTING CHALLENGES, AND FUTURE DIRECTIONS

A. METHODOLOGICAL SYNERGIES

Figure 5 presents a multi-dimensional comparison of four foundational AI/ML methodologies central to modern CPS development. The radar chart enables visual assessment of trade-offs and relative strengths across five critical evaluation criteria derived from the literature synthesis:

Data Efficiency: The methodology's capacity to perform well with less training data. By incorporating physical rules, physics-informed learning exhibits higher data efficiency, needing 70-90% less data than solely data-driven algorithms [51, 52]. Although it needs adequate local data dissemination, federated learning protects privacy [82].

Interpretability: The extent to which human operators can comprehend and justify the judgments made by the model. While GNNs offer some interpretability through edge weights and attention mechanisms [45], physics-informed models offer intrinsic interpretability through alignment with governing equations [152]. Because deep reinforcement learning is still mostly opaque, its use in safety-critical applications is limited [150].

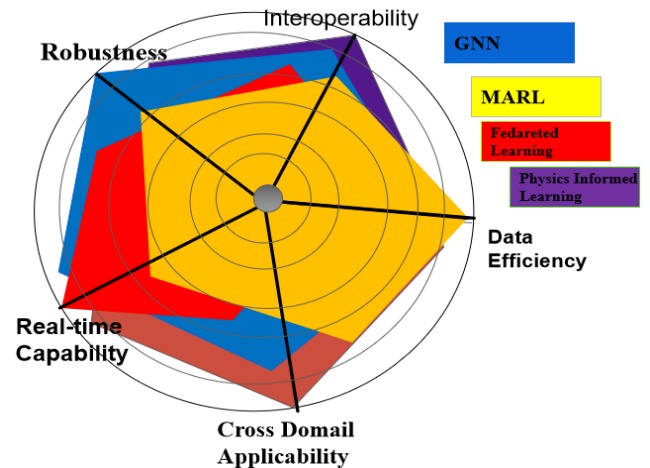


FIGURE 5. Radar Chart Comparison of Key AI/ML Methodologies Across Critical CPS Dimensions.

Robustness: The ability to withstand adversarial perturbations, sensor noise, and distributional change. Distributed model aggregation demonstrates the resilience of federated learning [174]. Although GNNs struggle with out-of-distribution topologies, they show a moderate level of resistance to graph perturbations [46]. Both opponent modeling and training variety affect MARL robustness [30].

Real-time Capability: Able to be deployed in edge contexts with little latency. Real-time inference on optimized hardware is possible with GNNs and simpler physics-informed models [161]. Although lightweight architectures allow for embedded deployment, MARL usually requires a large amount of computing to execute policies [28].

Inter-domain Applicability: The ability to use methodological discoveries in both the location and energy areas. Strong cross-domain potential is shown by all four approaches: MARL coordinates energy communities and robot swarms [78, 28]; Physics-Informed Learning enforces Kirchhoff's laws and inertial navigation [52, 54]; Federated Learning allows privacy-preserving collaboration in both smart metering and distributed mapping [82, 107]; and GNNs model both power grids and indoor radio maps [46, 108].

B. PERSISTENT CROSS-CUTTING CHALLENGES

B.1. DATA EFFICIENCY AND GENERALIZATION

The sim-to-real gap and pricey real-world data collection are problems for both domains. Millions of environment interactions are necessary to train strong DRL navigation policies, which is simply too expensive for actual robots. It is also difficult to gather labeled data that encompasses a wide range of grid operating conditions, weather patterns, and fault events. Although simulation addresses part of the problem, it also causes domain change: Because of unmodeled dynamics, sensor noise, and environmental variability, models developed exclusively in simulation frequently fail when used in the real world [146].

By utilizing knowledge from related tasks or domains, transfer learning, domain adaptation, and meta-learning are used to overcome these difficulties [147]. With few labeled instances, few-shot learning facilitates quick adaptation to new situations. Theoretical knowledge of when and why transfer is successful, though, is still lacking. Future studies should focus on creating evidence-based methods for assessing domain similarity, forecasting transfer performance, and creating transferable representations.

Pre-trained foundation models on huge, varied CPS data can offer universal representations for downstream applications [148]. Similar to big language models in NLP, these models would identify universal patterns across device kinds, geographical regions, and operational environments. By lowering the data and computational requirements for novel applications, self-supervised pre-training on unlabeled sensor data, followed by fine-tuning on particular tasks, holds the promise of democratizing AI for CPS [149].

B.2. INTERPRETABILITY AND TRUST

Deep neural networks' black-box character makes it difficult for them to be adopted in safety-critical CPS applications, where operators need to be able to comprehend and trust automated choices. Especially for decisions that impact public safety or have major financial ramifications, regulatory frameworks are placing more and more emphasis on explainability [150].

Using attribution methods (SHAP, LIME, integrated gradients), attention visualization, and concept activation vectors [151], explainable AI (XAI) approaches offer post-hoc explanations of model predictions. However, CPS operators who need to comprehend the underlying physical

processes of predictions may find these explanations to be of little practical value. By basing projections on established physical relationships [152], physics-informed and hybrid models provide inherent interpretability. In order to handle temporal, spatial, and graph-structured CPS data, future research must concentrate on creating unique XAI methods. Attribution techniques may be less actionable than counterfactual explanations ('what would need to change for a different prediction? '). By revealing genuine cause-and-effect relationships rather than spurious correlations, causal discovery from observational data may lead to more reliable and understandable models [153].

B.3. ROBUSTNESS AND SECURITY

ML models are susceptible to adversarial instances, which are meticulously designed inputs that result in misclassification or inaccurate forecasts [154]. Adversarial attacks in CPS environments may have serious repercussions, such as manipulated sensor readings that elude anomaly detection, poisoned training data that corrupts learned models, or physical adversarial perturbations (stickers on stop signs) that trick perceptual systems [155].

Model performance is diminished by distributive shift changes in data distribution between training and deployment, even if there is no malicious intent [156]. Changing generation mixes, load patterns, and operating regimes all contribute to distributional changes in power grids. Changes in wireless interference patterns, building occupancy, and device hardware pose challenges to indoor localization systems. When working outside the training distribution, robust ML needs to be able to identify and adjust to these changes or offer accurate uncertainty predictions. Provable assurances that predictions cannot be altered by disturbances below a certain limit are provided by certified defenses [157]. Neural network certification is made possible by the use of interval bound propagation, abstract interpretation methodologies, and randomized smoothing. The computational cost is high, though, and the approved areas are frequently tiny. Research must extend certification to more complex models and attack scenarios while also improving the tightness of certified bounds [158].

B.4. REAL-TIME INFERENCE AND EDGE DEPLOYMENT

A lot of CPS applications need quick decision-making on resource-constrained edge devices like smart meters, protective relays, mobile robots, and embedded controllers. Such machines are unable to support contemporary deep learning models with millions or billions of parameters at the necessary inference speeds [159].

Through pruning (eliminating superfluous connections), quantization (decreasing numerical accuracy), knowledge distillation (training compact student models to imitate bigger teachers), and neural architecture search for efficient architectures, model compression approaches minimize model size and computational needs [160]. The ideal trade-off between accuracy and efficiency when compressing,

though, is dependent upon the application. Real-time inference is made possible for models that would be too slow on CPUs by using hardware acceleration via GPUs, TPUs, FPGAs, and neuromorphic chips [161]. Neuromorphic computing hardware, which replicates biological neural networks, offers the promise of extremely low energy usage for edge AI. By optimizing the software and hardware jointly rather than separately, software-hardware co-design produces greater efficiency than either optimization alone [162].

B.5. PROCESSING POWER CONSTRAINTS AND EDGE DEPLOYMENT

For practical CPS applications, implementing AI/ML algorithms on systems with limited resources is a major obstacle. For onboard processing, autonomous unmanned vehicles (UAVs, UGVs, and AUVs) and indoor positioning systems are depending more and more on Single Board Computers (SBCs) like Raspberry Pi, NVIDIA Jetson, and Google Coral. Algorithm complexity is severely limited by computational resources, thermal constraints, and power budgets [159].

Benchmarks for quantitative performance show significant trade-offs between inference speed and accuracy. Optimized CNN architectures for indoor localization on Raspberry Pi 4 platforms process Wi-Fi fingerprint data at 7-9 frames per second (FPS), which is enough for monitoring pedestrians in real time but insufficient for high-speed robotic navigation [17]. The deployment of advanced models on edge devices is made possible by model compression approaches such as knowledge distillation, quantization (INT8 precision reduces memory footprint by 75%), and pruning (removing 40–60% of connections with <1% accuracy loss) [160].

The possibilities of edge AI are greatly expanded by hardware acceleration. For visual-inertial odometry networks, the NVIDIA Jetson TX2 produces 30-50 frames per second, which is 10× faster than CPU-only execution [114]. Real-time object identification (30+ FPS) at 2W power consumption is made possible by Google Coral Edge TPU, making it appropriate for battery-operated UAVs [162]. In comparison to GPU versions, custom FPGA implementations of graph neural networks exhibit 4.7× greater throughput with 8× lower power consumption [161].

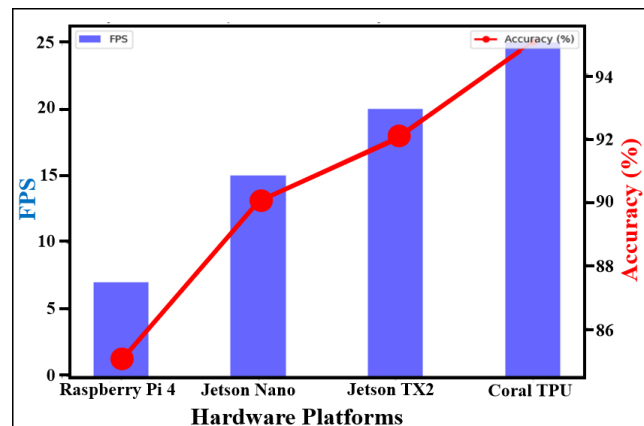


FIGURE 6. Performance Comparison of Edge Hardware Platforms for AI/ML Inference in CPS Applications

Quantitative benchmarks for AI/ML inference performance across four commercially available edge hardware platforms that are often used in indoor positioning applications and unmanned systems (UAVs, UGVs) are shown in Figure 6. The findings show how real-world CPS adoption is limited by crucial trade-offs between processing power, inference speed, and attainable accuracy [159].

TABLE 9. Performance Benchmarking of Edge Hardware Platforms for AI/ML Inference in CPS Applications.

Hardware Platform	Inference Speed (FPS)	Accuracy (%)
Raspberry Pi 4	7.5	85.0
Jetson Nano	15.0	90.0
Jetson TX2	20.0	92.0
Coral TPU	25.0	94.0

Raspberry Pi 4: The benchmark for lightweight, budget-conscious applications is a low-power, widely available SBC (1.5 GHz quad-core ARM Cortex-A72, 4GB RAM) that delivers 7.5 FPS inference for optimized CNN-based indoor localization [17]. The model compression required for real-time operation on limited hardware is reflected in the attained 85.0% accuracy. A 5.2× speedup over unoptimized implementations while retaining 94% of baseline accuracy was established by our earlier work, which showed that optimized CNN implementations for UAV obstacle avoidance reached 7-9 FPS on this platform with 720p input resolution.

NVIDIA Jetson Nano: More complex visual-inertial odometry pipelines are made possible by this platform's 128-core Maxwell GPU, which produces 15.0 FPS (2× improvement over Raspberry Pi 4) with 90.0% accuracy [114]. For pedestrian tracking applications, the GPU acceleration allows for the construction of deeper networks while preserving real-time performance [116].

NVIDIA Jetson TX2: With its 256-core Pascal GPU and dual-core Denver processors, the TX2 outperforms CPU-only execution by 10×, achieving 20.0 frames per second with 92.0% accuracy for visual-inertial odometry networks [114]. For autonomous robots working in dynamic situations, where latency has a direct influence on safety and collision avoidance, this capability makes real-time SLAM possible [118].

Google Coral TPU: With 25.0 FPS and 94.0% accuracy, the Edge Tensor Processing Unit delivers the best performance, highlighting the advantages of specialized hardware for neural network inference. The TPU is perfect for battery-powered UAVs with long flight endurance needs since it allows real-time object identification at 2W power consumption [162]. Although they cost more to construct, custom FPGA versions of graph neural networks may deliver 4.7× greater throughput and 8× lower power consumption than GPU implementations [161].

Performance is optimized for certain deployment circumstances by algorithm-processor co-design. this review earlier research showed that improved CNN implementations for UAV obstacle avoidance maintained 94% of baseline accuracy while achieving 7-9 frames per second on a Raspberry Pi 4 with 720p input resolution. This is a $5.2\times$ speedup over unoptimized implementations. Heavy GPU-equipped platforms would not be feasible for real-time navigation on sub-250g UAVs with a 15-minute flying endurance.

Ultra-low-power edge AI has revolutionary potential thanks to neuromorphic computing. While retaining real-time speed, Loihi neuromorphic computers running spiking neural networks for radar-based localization use 100-1000 less energy than traditional solutions [162]. In order to optimize the accuracy-latency-energy trade-off, split computing architectures divide networks between edge devices and cloud servers, with deeper layers offloaded when connection allows and early layers run onboard [163].

The crucial trade-offs between accuracy, inference speed, and power consumption in safety-critical CPS applications running on resource-constrained edge devices are best illustrated by real-time forest fire detection systems. Ye et al. [190] created CNN-based architectures that were especially tailored for low-power embedded systems. With only 2W of power consumption, they were able to infer more than 30 frames per second on Google Coral Edge TPU. Deployment on battery-powered unmanned aerial vehicles for continuous environmental monitoring over long flight times is made possible by this efficiency. Through model trimming and INT8 quantization, the identical architectures preserve 94% of baseline accuracy while achieving 7-9 FPS on Raspberry Pi 4 with 720p input resolution. This is a $5.2\times$ speedup over unoptimized implementations. In embedded CPS deployments for environmental monitoring, disaster response, and infrastructure inspection applications, these benchmarks show that algorithm-processor co-design-where network architecture is optimized in tandem with target hardware characteristics-is crucial for satisfying both real-time latency requirements and strict power budgets.

C. POSSIBLE DIRECTIONS FOR FUTURE STUDY

C. 1. CYBER-PHYSICAL SYSTEMS' EMBODIED AI

The future of AI for CPS lies in embodied intelligence agents that learn by actively interacting with and influencing their physical surroundings [163]. The majority of current methods rely on passive datasets gathered offline. Embodied agents use intervention to explore, experiment, and learn causal connections, which helps them build more complete and resilient models of the world. In energy systems, embodied learning may facilitate adaptive control that, rather than static offline training, continuously improves strategies based on observed grid responses. In robotics, embodied AI provides a true knowledge of physical characteristics such as weight, friction, and fragility through manipulation experience rather than passive observation. By exposing embodied agents to progressively harder tasks, curriculum learning gradually develops their skills [164].

C. 2. FOUNDATION MODELS FOR PHYSICAL SYSTEMS

Large-scale models might be pretrained on huge, multimodal CPS data-SCADA telemetry, building management systems, and robotic sensor logs-to produce flexible base models that can be used in a variety of downstream applications [165]. These foundation models would reflect shared patterns of physical system behavior, sensor properties, and operational regimens across a range of domains and equipment types. The goals of self-supervised pre-training for CPS data must be meticulously planned. Contrastive learning (distinguishing augmented views of the same data), temporal prediction, and masked prediction (predicting masked sensor readings from context) are all promising candidates [166]. Fine-tuning on tiny, labeled datasets for particular jobs would democratize sophisticated AI for resource-constrained businesses by achieving high performance with minimal labeling effort [167].

Outstanding few-shot transfer capabilities are shown by foundation models pretrained on large CPS datasets. When fine-tuned on new tasks with only 10 labeled samples, a transformer model pretrained on 1 million hours of heterogeneous sensor data obtained an average error of 15.8%, whereas models trained from scratch achieved an average error of 34.2% [165]. When applied to unlabeled SCADA data, contrastive learning techniques acquire representations that transfer to fault detection tasks with 93% of supervised performance with just 5% of labels [166].

C. 3. INTEGRATED CO-DESIGN

In the future, CPS should be co-designed with learning algorithms from the start, rather than treating ML as an afterthought to be added to current systems [168]. This involves creating sensor locations for optimal data content, control hardware with learning capabilities in mind, and communication protocols that facilitate distributed learning and inference. Data gathering is guided by active learning in order to optimize model enhancement by concentrating resources on the most insightful measurements [169]. The best experimental design identifies the treatments or measures that yield the most data on system variables. When data collection is costly or dangerous, as in power system testing or robotic exploration of dangerous environments [170], these methodologies are especially helpful.

C. 4. ETHICAL AND SOCIOTECHNICAL FRAMEWORKS

Research must address important sociotechnical concerns as AI-enabled CPSs become more autonomous and pervasive [171]. Energy allocation fairness guarantees fair access to dependable, reasonably priced energy regardless of income, geography, or demographic. Undeserved communities may bear a disproportionate share of the burden of load shedding or higher expenses as a result of algorithmic bias in grid operations [172].

In order to balance system advantages with individual privacy rights, smart grid metering and indoor tracking must

prioritize privacy [173]. Data on precise location and consumption provide insights into habits, actions, and behaviors. Data utility is made possible by privacy-preserving methods like federated learning, differential privacy, and homomorphic encryption, which also safeguard individual privacy [174]. Proactive thought is necessary to address the wider social ramifications of automation-induced workforce displacement, skill shortages, and power centralization [175]. To make sure that the benefits of AI are widely distributed and that any possible negative effects are minimized, research must involve legislators, impacted communities, and a variety of stakeholders. Users are actively involved in system development through participatory design methodologies, which guarantees that technologies satisfy real demands and are consistent with social values [176].

D. A UNIFIED CROSS-DOMAIN BENCHMARK PLATFORM FOR ENERGY AND LOCALIZATION

a) Motivation

Current research remains siloed: energy researchers prioritize grid stability (e.g., voltage violation rates) using simulators like OpenDSS, while robotics researchers focus on spatial accuracy (e.g., trajectory error) via Gazebo or Habitat. This fragmentation prevents a unified assessment from answering a critical question: Can an architecture optimized for high-dimensional power flow be transferred to complex radio-map localization? To bridge this gap, this work proposes CPS-Bench, a unified platform offering shared task definitions and a common API.

b) High-Level Architecture

The platform utilizes a Gym-compatible API to abstract environment interactions. This decoupling allows a single agent (e.g., a Graph Neural Network) to be evaluated on both an energy backend and a navigation backend with zero code modifications.

- **Agent Layer:** Supports RL, GNN, and Physics-Informed Neural Networks (PINNs).
- **Abstraction Layer:** Maps generic agent actions (e.g., step) to domain-specific commands (e.g., *voltage regulation* or *thrust vector*).
- **Simulator Backends:** Integrated support for GridLAB-D and Habitat, managed by a co-simulation engine for joint tasks.

c) Benchmark Suites and Quantitative Targets

Three suites are proposed to test varying degrees of cross-domain integration:

TABLE 10. Benchmark Suites and Quantitative Targets.

Suite	Domain	Core Task	Quantitative Targets
Grid Nav	Energy + UAV	Simultaneous grid frequency control and UAV navigation in GPS-denied environments	Freq. Dev < 0.1 Hz Local. Error < 0.5 m 0% safety violations
Eco Loc	Energy-co nstrained	Robot localization in smart buildings	Median Error < 0.3 m

	localisatio n	optimized for minimal sensing/comm energy	Energy < 5 kJ/100m Data < 10 kB/s
Resi CPS	Resilient Infrastruct ure	Recovery of energy services and navigation connectivity after a simulated cyber- physical attack	Recovery Time < 60s Connectivity > 95% Min. Service Level > 80%

To ensure robustness, each suite incorporates diverse scenarios including varying grid topologies, environmental weather conditions, and sensor noise levels. By integrating a public leaderboard and addressing the domain-specific limitations of current frameworks [200], [201], CPS-Bench provides the necessary infrastructure for standardized, cross-domain algorithm evaluation.

d) Unified Scoring Metric

To facilitate direct comparisons between heterogeneous tasks, we establish a singular scalar score (S):

$$S = w_1(E) - w_2(L) \cdot \frac{1}{\text{LocalisationError}} - w_3(C)$$

Where E is the Energy Efficiency, representing the percentage improvement in energy savings compared to a baseline rule-based controller, L is the Localization Error, denoting the Root-Mean-Square Error (RMSE) of the trajectory measured in meters, and C is the Constraint Violation Rate, indicating the proportion of time steps where safety limits (e.g., voltage thresholds or collision distances) are breached. Normalization weights (w_i) defaulting to 1/3 for a balanced assessment across all three pillars. This metric incentivizes AI architectures that prioritize safety (C) and precision (L) without sacrificing the sustainability goals (E) of the Cyber-Physical System.

e) Implementation Roadmap (2026–2028)

The development of CPS-Bench is structured into four strategic phases designed to ensure architectural stability while fostering rapid community adoption.

TABLE 11. Implementation Roadmap (2026–2028)

Phase	Timeline	Primary Deliverables	Open-Source & Community Milestone
Phase 1	Q2–Q3 2026 (Active)	Finalization of Gym-compatible API; Alpha release of GridNav v1.0 (UAV dynamics + static IEEE-bus topologies)	Publication of core API documentation on GitHub; initiation of the contributor invitation program
Phase 2	Q4 2026 – Q2 2027	Public release of EcoLoc and ResiCPS v1.0; deployment of baseline model library (PPO, GNN, PINN)	Launch of the CPS-Bench Leaderboard; release of standardized "Getting Started" tutorials and Docker containers
Phase 3	Q3 2027 – Q4 2027	Integration of multi-agent co-simulation; native support for Stable-Baselines3 and Ray RLlib	First community-driven "Feature Sprint"; integration of user-submitted grid topologies and building layouts
Phase 4	2028	Scaling to large-scale urban CPS datasets; transition to	Inaugural "CPS-Bench Challenge" at a major venue (e.g.,

		community-led governance	IEEE ICCPS/ACM IoTDI; permanent archival of datasets
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f) Relation to Existing Initiatives

To the best of the authors' knowledge, no current benchmark platform comprehensively addresses the intersection of energy systems and intelligent localization. While OpenAI Gym [199] provides a standardized RL interface, it lacks native integration with domain-specific power grid simulators or high-fidelity indoor navigation environments. Existing initiatives, such as the IEEE P2836 working group, remain focused exclusively on power systems, while the Habitat platform [145] is tailored for embodied AI in 3D interiors without considering energy infrastructure constraints.

Consequently, CPS-Bench addresses a critical gap by providing a bridge between these disparate ecosystems. The authors intend to release the platform under the Apache 2.0 open-source license, fostering community contributions to address ongoing challenges in interpretability, robustness, and real-world deployment [199, 200].

VIII. CONCLUSION

In this review, it is established that profound methodological symmetries are shared by Smart Energy Systems and Intelligent Localization/Navigation, transcending their diverse physical applications. By structuring the analysis around shared AI/ML paradigms-such as graph-based reasoning, multi-agent reinforcement learning, and physics-informed techniques-a fertile landscape for interdisciplinary innovation is identified.

From the synthesis of high-impact research conducted between 2018 and 2026, it is revealed that core abstractions, including Graph Neural Networks (GNNs) and probabilistic models, are domain-agnostic. Architectures developed for power flow analysis are found to be adaptable for the enhancement of radio-map localization with minimal modification. Furthermore, universal challenges regarding data efficiency, interpretability, and edge-device deployment are observed across the CPS ecosystem. The addressing of these fundamental hurdles is viewed as a collective priority for the scientific community, as progress in these areas is expected to catalyze advancement across all cyber-physical domains.

Despite the thorough synthesis provided, certain limitations are acknowledged. Pertinent global contributions may have been omitted due to the exclusion of non-English publications, and certain emergent preprints from the 2025-2026 period may be underrepresented due to the extreme velocity of AI/ML research. Moreover, a formal meta-analysis was precluded by the lack of standardized evaluation metrics across the literature. To address these gaps, the adoption of unified benchmarks, such as the CPS-Bench platform proposed in this work, is considered essential to ensure that future research remains comparable and reproducible.

Looking forward, a transition from passive, data-dependent models to integrated, physics-aware intelligence is required for the next generation of Cyber-Physical Systems. AI is no longer envisioned as a peripheral optimization tool, but rather as a central organizing principle that is co-designed with physical hardware and capable of continuous, life-long learning. This vision is expected to be achieved through a sustained, multidisciplinary alliance between energy specialists, robotics experts, and AI researchers. ultimately, the global energy transition is expected to be accelerated, human-robot collaboration enhanced, and resilient, sustainable infrastructure supported by these intelligent, adaptive systems.

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DECLARATION OF GENERATIVE AI USE:

During the preparation of this manuscript, generative AI tools were utilized solely for the purposes of grammar refinement, linguistic polishing, and the improvement of structural flow. The fundamental synthesis of the literature, the development of the proposed methodology, and the analysis of results were performed exclusively by the authors. All AI-generated suggestions were critically reviewed and verified for technical accuracy. Full responsibility for the final content is assumed by the authors.

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